

Guidelines for robust and reproducible point defect simulations in crystals

Alexander G. Squires¹, Seán R. Kavanagh², Aron Walsh³ & David O. Scanlon¹

Abstract

Many physical properties of functional materials are governed by their impurities rather than their bulk characteristics. Defects in crystals can activate electronic and ionic conductivity, create active centres for catalysis or store information through localized spin configurations. Accurate modelling of defect behaviour is therefore essential for predicting material performance and optimizing functionality across a vast application space. However, defect simulations are sensitive to choices made during setup, execution and analysis. In this Perspective, we highlight best practices for calculating and reporting point defect properties through computational methods, with a focus on the widely adopted supercell approach. Key considerations include accurate representation of the structural and electronic properties of the host material, appropriate choice of charge states, sufficient optimization of defect geometries and reproducible calculation of defect formation energies. Adhering to these practices will facilitate robust comparisons between studies and improve the integration of computational predictions with experimental results. We emphasize the importance of reporting computational parameters and correction schemes. Ultimately, an open approach to point defect simulations will strengthen the impact of computational studies and accelerate materials engineering.

Sections

Introduction

The host material

Electronic structure

Dielectric screening

Chemical potentials and phase stability

Selecting defects and their charge states

Defect structure searching

Simulating the defective cell

Addressing the limitations of the supercell method

Quantifying defect populations

Practical steps to reproducible defect analysis

¹School of Chemistry, University of Birmingham Edgbaston, Birmingham, UK. ²Yusuf Hamied Department of Chemistry, University of Cambridge, Cambridge, UK. ³Department of Materials, Imperial College London, London, UK. ✉ e-mail: a.squires@bham.ac.uk; sk2045@cam.ac.uk; a.walsh@imperial.ac.uk; d.o.scanlon@bham.ac.uk

Introduction

Defects are intrinsic to extended solids, with an equilibrium population arising from entropic contributions to free energy¹. Acting as sources and sinks for charge carriers, defects profoundly affect mechanical, electronic and transport properties of a material, influencing applications including photovoltaics^{2–6}, batteries^{7–13}, quantum computing^{14,15}, transparent conductors^{16–18}, power electronics^{19,20} and beyond²¹.

In this Perspective, we consider best practices and improved methodological reporting for computational studies of point defects, such as vacancies, interstitials and substitutions, whether intrinsic or extrinsic dopants. We note that extended defects such as grain boundaries and interfaces introduce additional structural and electronic complexity, which can influence charge redistribution, recombination pathways and local band alignment. The considerations they raise are numerous, however, and here we maintain a focus on point defects. We outline methodological considerations affecting calculation accuracy, highlighting common pitfalls, and provide guidelines for transparent reporting. Clear standards enhance reproducibility and reduce redundant efforts and are becoming especially crucial as computational defect databases emerge^{22–24}, and machine-learning models increasingly leverage these data sets^{25–28}. The quality of such predictive models relies fundamentally on the accuracy of the underlying computational data, following the common adage; ‘garbage in, garbage out’ or, ideally, ‘quality in, quality out’.

The impact of a defect species on a material property is a function of its concentration. Accurately quantifying defect populations

is therefore essential for understanding their influence on material properties, as defects do not form with equal probability. Defect concentrations are often extremely low, but concentrations of one-in-a-million (10^{16} cm^{-3}) or lower can make-or-break certain applications; for example, 0.1 ppm titanium contamination in silicon can cause a ~2% loss in solar cell efficiency²⁹. In high-purity chemical vapour deposition diamond, introducing just 0.2–20 ppm nitrogen results in the formation of nitrogen-vacancy centres at concentrations as low as the parts-per-billion level, and these 10^{-9} – 10^{-8} fractional populations dominate the optical and spin behaviour of the material³⁰. These low concentrations make experimental observation and manipulation difficult owing to limitations in spatial and temporal resolution. Atomistic computational methods are thus indispensable, providing a means to model and predict point defect behaviour from first principles.

Such low concentrations are treatable within the dilute limit, within which the equilibrium concentration of a defect X is given by:

$$[X] = N_X \exp\left(\frac{-\Delta G_f^X}{k_B T}\right), \quad (1)$$

in which N_X is the concentration of sites upon which defect X can form and ΔG_f^X is the free energy of formation of the isolated point defect X (refs. 31–33). Owing to the inverse exponential relationship between defect formation energy and concentration, small shifts in formation energies – from changes in material growth conditions, for instance – can drastically affect defect concentrations, as illustrated in Fig. 1a,b.

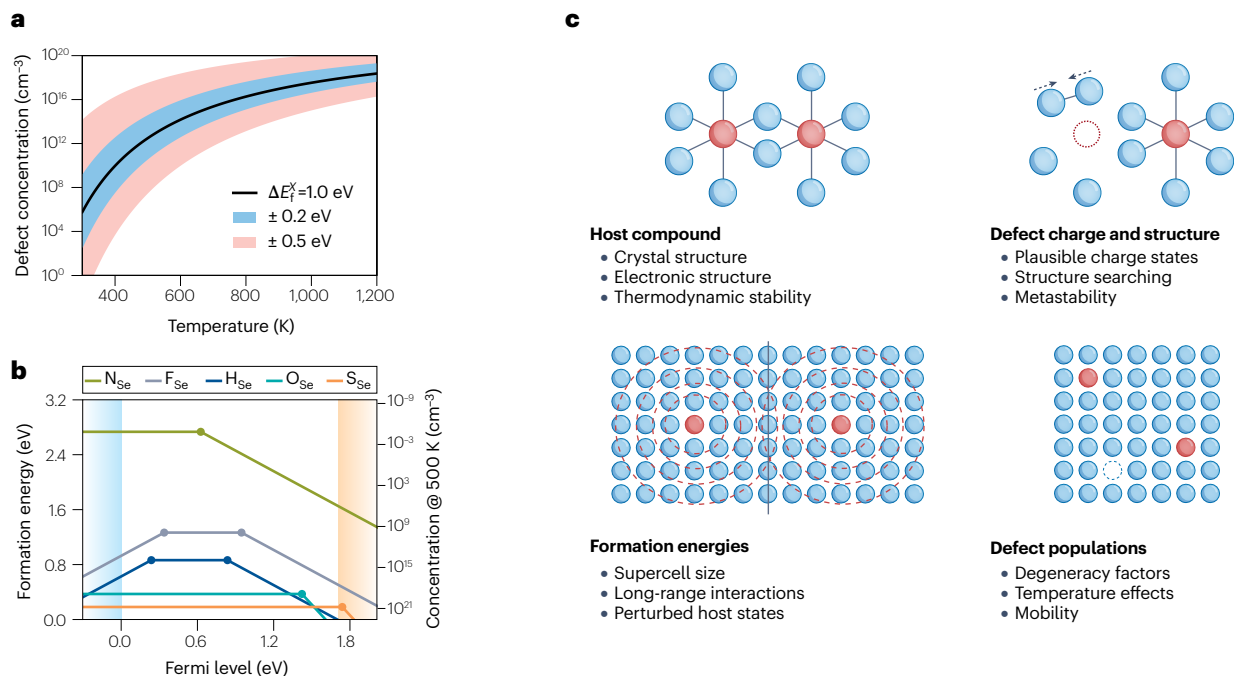
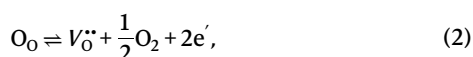


Fig. 1 | Considerations for accurate defect modelling. **a**, Equilibrium defect concentration as a function of temperature for a defect X with a formation energy $\Delta E_f^X = 1 \text{ eV}$ and site multiplicity N_X of $3.7 \times 10^{22} \text{ cm}^{-3}$. Variations in concentration arising from differences or errors in the formation energy of $\pm 0.2 \text{ eV}$ and $\pm 0.5 \text{ eV}$ are shown in blue and pink. **b**, Calculated formation energy diagram for extrinsic substitutions (X_{Se}) in trigonal selenium⁶, plotting charge-dependent defect formation energies and corresponding equilibrium concentrations at 500 K as a function of Fermi-level position. The valence band maximum is set to 0 eV. Blue shading represents the valence band and orange shading represents the

conduction band. A guide to interpreting these diagrams can be found in ref. 179. **c**, The first step in a computational defect study involves understanding the host material, including its local atomic structure, electronic structure and thermodynamic stability. Next is the investigation of defect charge states and structures, focusing on accessible charge states, (global) optimization of geometries and the presence of metastable states. Defect formation energies are then calculated, which should account for periodic image interaction and any perturbed host states. Finally, defect populations can be quantified, considering configurational degeneracies, temperature effects and defect diffusion.

Likewise, small differences in computational parameters or approaches can greatly alter predicted defect populations. To approach quantitative accuracy in predicting defect populations and thus characterizing their impact on properties of interest, it is essential to carefully calculate defect formation energies. Moreover, transparent reporting is crucial, allowing any potential sources of error or discrepancies between studies to be meaningfully evaluated.

The central procedure for calculating the defect formation energy is through the evaluation of a Hess cycle. For example, consider the formation of an oxygen vacancy in an oxide:



in which Kröger–Vink notation³⁴ is used, such that O_0 denotes an oxygen atom on an oxygen lattice site, and $\text{V}_0^{\bullet\bullet}$ indicates a vacant lattice site which has a +2 charge relative to O_0 in the pristine (bulk) structure – henceforth denoted simply as V_0^{2+} .

As with any chemical reaction, the formation energy is given by:

$$\Delta E_f = \sum_{\text{products}} n_i E_i - \sum_{\text{reactants}} n_i E_i \quad (3)$$

$$\rightarrow \Delta E_f^{\text{V}_0^{2+}} = [E(\text{V}_0^{2+}) + \frac{1}{2}E(\text{O}_2) + 2E(\text{e}^-)] - E(\text{O}_0). \quad (4)$$

Here, free energies have been approximated by internal energies, which are more readily computed. For a general point defect, the net formation energy can be expressed as:

$$\Delta E_f^{X^q} = E^{X^q} - E^{\text{bulk}} - \sum_i n_i \mu_i + qE_f. \quad (5)$$

In this equation, the reactant energies are the pristine bulk material, E^{bulk} , and any atoms ($n \geq 1$) or electrons ($q \leq 1$) added to the system. The product energies are the defective system, E^{X^q} , and any removed atoms ($n \leq 1$) or electrons ($q \geq 1$). Atoms and electrons are exchanged with external reservoirs, in which the atomic chemical potentials, μ_i , describe the energies for atoms, and the Fermi level, E_f , accounts for the electrons. For the oxygen vacancy example in equation (2), these correspond to the energies of releasing half an oxygen molecule and two electrons, respectively^{31–33,35}, such that $E(\text{V}_0^{2+})$ corresponds to E^{X^q} , $\frac{1}{2}E(\text{O}_2)$ to $-\sum_i n_i \mu_i$, $E(2\text{e}^-)$ to qE_f and $E(\text{O}_0)$ to E^{bulk} .

Computational methods allow, in principle, accurate calculation of each of the terms entering the defect formation energy, enabling quantitative prediction of $E_f^{X^q}$ and of the resulting defect concentrations $[X]$ ³². Although conceptually straightforward, accurate evaluation of equation (5) entails a range of technical challenges. Even when these are resolved, the underlying modelling framework rests on a set of simplifying assumptions: most studies adopt a thermodynamic formalism based on internal energies, dilute non-interacting defects and equilibrium statistical mechanics. Although this approach incorporates configurational entropy (via equation (1)) and can account for other entropic terms (such as spin or orientation) through a degeneracy pre-factor, it often omits other finite-temperature effects³⁶. Approximations are necessary to maintain computational tractability, but they impose limitations. As such, careful interpretation of predicted concentrations – and explicit acknowledgement of the assumptions and uncertainties in the model – is essential. Experimentally validating computational defect predictions requires techniques that are sensitive to low concentrations, such as electron spin resonance or deep-level transient spectroscopy, and often their use

Table 1 | Software for pre-processing and post-processing of point defect calculations

Package	Description	Licence	Ref.
doped	Comprehensive defect simulation toolkit	MIT	85
pydefect	Defect generation and analysis, emphasis on high throughput	MIT	25
spinney	Post-processing of first-principles defect calculations	MIT	113
AiiDA-defects	Defect calculation workflows in the AiiDA ecosystem	MIT	177
pmg-analysis-defects	Defect generation and analysis tools within pymatgen	BSD	105
defap	Analysis of defect chemistry and thermodynamics	BSD	153
pylada-defects	Defect generation and processing, emphasis on high throughput	LGPL	101
PyCDT	Tools for high-throughput defect calculations	LGPL	102
pydecs	Workflow for analysis of point-defect densities in crystalline solids	Apache	178
DASP	Calculating defect and dopant properties in semiconductors	Closed source	112

as a function of growth conditions to confidently assign behaviour to a particular species.

To support reproducibility, comprehensive reporting of computational parameters, software versions and workflows is essential, aligned with FAIR principles³⁷. Open-source software (Table 1) facilitates automation and analysis, but variations in implementation choices and default settings still necessitate detailed documentation. Many of the methodological considerations discussed herein apply broadly to different defect-modelling strategies, yet our focus is on the widely used supercell approach^{32,33,38}. In this framework, the defect is embedded in a periodically repeated host to approximate dilute concentrations, leveraging the maturity and robustness of periodic density functional theory (DFT) codes, which typically provide reliable bulk descriptions with modest technical overhead.

Although plane-wave basis sets are commonly used in such calculations, alternative localized-orbital approaches are also used. These offer advantages such as compact basis sets, efficient hybrid-functional scaling and flexibility for non-periodic or embedded geometries. However, their reliance on atom-centred basis functions introduces basis-set bias, and vacancy-centre or colour-centre states may require augmentation with diffuse or ‘ghost’ functions. The placement and composition of these functions are not unique and can influence localization, symmetry and defect-level positions, introducing a degree of user dependence absent in plane-wave formulations. These factors do not preclude alternative localized-orbital-based defect studies, but they underline the importance of careful basis design and transparent reporting.

Supercell methods also sit within a broader landscape of defect-modelling approaches. Embedding schemes treat the defect and its local coordination environment at higher resolution while representing the surrounding material more approximately, using continuum or lower-level atomistic descriptions^{39–41}. These approaches avoid periodic images and can efficiently capture long-range elastic and electrostatic response, although their accuracy depends on how the outer

region is constructed and on the stability of the interface between regions of differing resolutions. The increasing availability of accurate machine-learned interatomic potentials offers an opportunity to build more realistic atomistic environments for such embeddings⁴¹, potentially extending their applicability to cases in which fully periodic treatments remain costly. Although these alternative strategies are valuable, the supercell approach remains the most widely adopted across the field, and we therefore centre our discussion on this methodology.

We categorize the key considerations and potential pitfalls of defect simulation workflows into several areas, outlined in Fig. 1c, which guide our discussion. These include accurately describing the host system, identifying stable defect structures and charge states, appropriately handling calculation outputs and formation energies and accurately calculating defect concentrations. Experienced defect practitioners will be aware of these issues, but detailed discussions remain sparse in the literature despite their influence on predictions.

The host material

The accuracy of defect formation energies is fundamentally limited by the description of the host material. Defects are typically defined relative to the host in a quasiparticle-like picture, where they are treated as perturbations (excitations) within an otherwise pristine atomic and electronic structure⁴². Computationally, this involves embedding the defect within the host crystal structure and calculating energies, forces and other properties to simulate dilute defect behaviour. It is essential that the host material is accurately described by the chosen computational methodology, as errors or instabilities in the underlying structure propagate through to the calculated defect energetics.

Defect formation energies are sensitive to both local and global atomic geometries owing to their influence on bonding and strain around the defect site⁴³. For example, van der Waals interactions are crucial for predicting defect stability and mobility in certain non-close-packed or layered materials, through their impacts on lattice parameters and bond lengths⁴⁴. This structural sensitivity, particularly the dependence of formation energies (and thus defect concentrations) on strain, has even been exploited in the so-called strain-engineering approaches to tune defect populations and properties^{45–47}. An additional source of error arises when the host crystal structure is relaxed using one computational setup, but defect calculations are performed with another⁴⁸. This inconsistency introduces artificial strain, as the host structure used in the defect calculation – relaxed, for example, using a semi-local functional such as PBE – is no longer at its equilibrium geometry when subsequently modelled with a different setup, such as a hybrid functional (for example, HSE06). Consequently, the defect-containing supercell experiences residual internal forces that would, were it able to relax, drive it towards the geometry that would be obtained with the hybrid functional. This mismatch leads to errors in the calculated relative energies, potentially giving misleading results for defect formation energies and predicted stability.

Another issue arises when the host structure is metastable within the chosen computational framework. If a defect breaks symmetry locally and triggers relaxation towards an alternative, lower-energy phase, it can induce a spontaneous phase transition throughout the simulation supercell^{28,36,49,50}. Such behaviour frequently occurs in materials exhibiting temperature-driven structural transitions, notably perovskites. For instance, many perovskites possess a cubic structure at room temperature but adopt lower-symmetry distortions (octahedral tilting and ion displacements) at low temperatures or under

static modelling conditions^{51–53}. Defect calculations initiated from a higher-symmetry (such as cubic) reference structure can thus relax into distorted, energetically favourable geometries, yielding defect formation energies that appear artificially low or even negative. In other words, the calculated defect formation energies reflect the energetics of a phase transition induced by the defect calculation rather than the intrinsic stability of the defect itself, leading to physically unrealistic interpretations and a lack of convergence with increasing supercell size.

To avoid such artefacts, it is essential to verify the (dynamic) stability of the host structure before conducting defect calculations. In cases of discrepancies between the experimental and athermal DFT structures (owing to finite-temperature effects, for example), there are several common approaches to attempt to represent the host structure. Each of these has trade-offs in terms of accuracy and computational cost:

- Use the athermal ground-state structure. Using the minimum in the potential energy surface (PES) is the safest option and serves as a consistent baseline. Although it may differ from the operating-temperature phase, trends are often transferable if polymorphs are structurally similar. For perovskite crystals, this may be a low-temperature orthorhombic polymorph, for example.
- Constrain to the experimental structure. Techniques such as the use of supercells that are incommensurate with certain symmetry-breaking distortions can suppress phase transitions during relaxation⁵⁴. However, these constraints may artificially inhibit local defect-induced distortions. For instance, an odd supercell expansion (such as $3 \times 3 \times 3$) can suppress rigid octahedral tilts in perovskites.
- Run finite-temperature simulations. Ab initio molecular dynamics allows direct sampling of equilibrium configurations but is computationally expensive. The average cubic phase of a perovskite would be well represented in ensembles from such simulations. Machine-learned force fields offer a promising route to lower the computational burden, although they remain under development and system-dependent^{28,55–57}.

Each approach has trade-offs in realism and computational cost. Unless justified otherwise, the athermal DFT structure should be used as a well-defined and reproducible reference. Regardless of the method, the host structure used in defect calculations must be clearly documented. This includes reporting the crystal structure, the computational setup used for relaxation and whether dynamic structural stability was assessed. If different levels of theory are used to characterize the atomic and electronic structure, the rationale and possible implications of this inconsistency must be addressed explicitly. It should be stated whether the relaxed structure corresponds to a ground-state polymorph, a higher-temperature phase or a constrained geometry, and whether structural constraints were imposed during relaxation. In systems with known-phase competition or transitions, any observed defect-induced structural changes should be noted and critically evaluated.

Electronic structure

A critical parameter in DFT is the choice of exchange-correlation functional, in which the quantum-mechanical interaction between electrons is treated. Semi-local functionals predict many bulk crystal structures and ground-state properties with reasonable accuracy, yet systematically underestimate band gaps and charge localization, which poses challenges in defect studies. Defect energy levels are defined

relative to the band edges, and mischaracterizing these leads to incorrect charge transition levels and carrier concentrations^{32,58–60}. Such errors can misclassify defects as donors, acceptors or recombination centres, affecting properties such as carrier lifetimes and luminescence. An accurate host band gap is therefore essential to describe defect behaviour correctly.

Hybrid functionals such as HSE06 and PBE0 mitigate this band gap error by incorporating a fraction of exact Hartree–Fock exchange^{32,61,62}. This fraction is often tuned to reproduce the experimental gap, but this introduces an empirical dependence and, without additional considerations, overlooks other key contributors such as spin–orbit coupling (SOC), disorder, temperature and electron–phonon interactions. For instance, early defect studies of CdTe omitted SOC corrections because HSE06 with 25% exchange adequately reproduced the experimental gap; however, SOC raises the valence band maximum by ~ 0.3 eV. Neglecting this led to predictions of strong donor compensation inconsistent with experimentally established p-type behaviour of CdTe⁶³. Although the calculated band gap appeared accurate, this seemingly successful result masked the underlying methodological inconsistency, exemplifying how obtaining the ‘right value for the wrong reason’ can obscure the actual source of error. Including SOC and subsequently adjusting the exchange fraction correctly reproduce both the experimental gap and the observed doping limits^{62,64,65}, highlighting the importance of rigorously assessing all significant contributions to electronic structure.

Koopmans-compliant hybrid functionals provide an alternative. These enforce piecewise linearity of the total energy with respect to fractional charge, aligning the frontier Kohn–Sham eigenvalues with ionization potentials and electron affinities⁶⁶. By tuning the fraction of Hartree–Fock exchange to satisfy the Koopmans condition, it is argued that these methods improve the positions of defect eigenvalues relative to the band edges⁶⁷. One limitation is that the optimal exchange may vary between different defects in the same host material⁶⁸. Dielectric-dependent hybrid functionals provide another strategy by relating the exact exchange fraction to the inverse of the high-frequency dielectric constant ($\alpha = \epsilon_\infty^{-1}$)^{69,70} of the bulk material. Although band gaps are generally improved following this strategy, the analysis is performed for the host materials and therefore may not improve the description of defects.

Post hoc correction schemes, such as scissor operators or *GW* eigenvalue shifts, adjust the band edges of the host. These can improve agreement with experimental band gaps but introduce inconsistencies when computing formation energies and charge transitions^{59,71,72}. Although *GW* can capture one-particle excitations, it is often used non-self-consistently and remains computationally prohibitive for large supercells. These methods partially correct the electronic structure but do not provide a self-consistent framework for defect energetics.

We briefly note the use of DFT + *U* as an inexpensive approach to increase localization for selected orbitals. Although it shares the same broad aim of improving the description of electronic structure⁷³, its limitations are more severe than those of hybrid functionals. The method requires an explicit partitioning of the electronic structure into correlated and uncorrelated subspaces, defined by user-chosen projectors or local orbital bases. Different choices of this partitioning can lead to different occupations, degrees of localization and defect-level positions⁷⁴. Together, these ambiguities render DFT + *U* highly sensitive to user choices. By contrast, hybrid functionals modify the exchange–correlation treatment in a uniform manner across the

system, yielding a more internally consistent and predictable description of the host electronic structure, even though they do not constitute a ground truth. DFT + *U* should therefore be regarded as an approximate and non-transferable correction, useful in some contexts but requiring careful justification and transparent reporting when applied to defect studies, although extensions to this methodology may improve its applicability to defect modelling^{75,76}.

In all cases, it is critical to report the approach used to model the host electronic structure. This includes the choice of DFT functional, the fraction of exact exchange used (if applicable), the inclusion or omission of SOC and any corrections or shifts applied to the eigenvalues or band edges. If any band gap tuning is performed, the procedure and justification should be stated. These methodological choices can lead to qualitatively incorrect conclusions if not handled carefully. More broadly, approaches that seek to enforce Koopmans compliance – whether achieved through a Hubbard *U*, a tuned hybrid, a dielectric-dependent exchange fraction or explicit Koopmans-corrected formalisms – can improve internal consistency in the frontier eigenvalues, but no practical realization should be regarded as a ground truth. The resulting electronic structures remain parameter-dependent and non-unique, and apparent improvements may reflect compensating errors rather than a systematically more accurate description. Clear reporting and careful interpretation are therefore essential when using any such tuning strategy.

Dielectric screening

Dielectric screening has a critical role in the behaviour of charges in materials. It has direct relevance to defect studies through its role in modulating (periodic) charge interactions and thus influencing finite-size corrections inherent to the supercell approach^{77,78}.

Dielectric screening can be described by rank-two tensors, which, for isotropic crystals, reduce to a single scalar value. It can be decomposed into two distinct components: $\epsilon_0 = \epsilon_\infty + \epsilon_{\text{ion}}$. The high-frequency (electronic) response, ϵ_∞ , reflects the ability to screen fast electronic (such as optical) perturbations; and the low-frequency (ionic) response, ϵ_{ion} , accounts for lattice polarization owing to ionic displacements. For the thermodynamics of defect formation, the full static response is usually the most relevant quantity, but for optical transitions, the high-frequency response has an important role⁷⁹.

These two contributions are typically computed using different methods owing to their disparate timescales. The ionic (low-frequency) dielectric response is usually calculated via density functional perturbation theory or finite-difference methods. Semi-local DFT functionals are often used here as they typically show good accuracies for lattice dynamics, whereas hybrid and beyond-DFT approaches tend to have high computational costs within these approaches. However, caution is required for materials with narrow band gaps, where the application of perturbations can induce spurious metallicity, leading to unphysical screening. By contrast, the electronic (high-frequency) dielectric constant is much more sensitive to the underlying electronic structure. Roughly speaking, $\epsilon_\infty \propto 1/\sqrt{E_g}$, and so semi-local DFT tends to overestimate ϵ_∞ owing to self-interaction errors and band gap underestimation. Methods that accurately model the host band structure should therefore be used (ideally using the same computational setup for host electronic structure, dielectric properties and defect calculations).

If different functionals are used to evaluate the electronic and ionic contributions separately, it is essential that both calculations are performed on geometries relaxed using the same functional. Mixing structures across functionals can introduce spurious strain

effects (as discussed earlier) that propagate into the calculated dielectric properties. In addition, dielectric predictions typically require tighter convergence settings than total-energy calculations, including more complete basis sets, stricter force thresholds for ϵ_{ion} and finer Brillouin-zone sampling for ϵ_{∞} (ref. 80).

To ensure accurate and reproducible predictions, all computational parameters related to dielectric screening should be explicitly reported. This includes the methods used to obtain ϵ_{∞} and ϵ_{ion} , the exchange-correlation functional, the structural geometry used and the key convergence settings such as basis-set completeness, k -point sampling and force criteria. Where different methods or functionals are used for the electronic and ionic responses, it should be stated clearly whether the calculations were performed on a consistent structure. Any known numerical sensitivities – such as spurious metallicity in narrow-gap systems – should also be noted, together with any mitigation strategies.

Chemical potentials and phase stability

Defect formation energies depend on the atomic chemical potentials of the product and reactant elements (equation (5)). These chemical potentials represent the thermodynamic conditions under which defects form – such as synthesis or processing environments – and determine how defect populations vary with changes in those conditions. For example, ‘ X -rich’ conditions correspond to a higher value of μ_X , which suppresses the formation of X vacancies and favours the formation of X interstitials/substitutions. However, the chemical potentials of the constituent elements are not free parameters. They are thermodynamically constrained to ensure that defect formation energies are referenced to a consistent and physically meaningful environment, typically one in which the host material is stable. Outside this stability region, the host may spontaneously decompose into competing compounds or elemental phases. In such cases, defect formation in the target host is no longer a well-defined thermodynamic process, as the material itself is unstable and does not provide a reliable energetic reference. Although such scenarios can be relevant for metastable materials, the standard approach in defect modelling is to restrict chemical potentials to the stability region of the host. This region is determined by solving a set of thermodynamic inequalities: the chemical potentials must reproduce the host formation energy, lie below their elemental reference values and not favour the formation of any competing phases⁸¹.

For example, in the case of a binary oxide A_xO_y , the allowed chemical potentials must satisfy:

$$\mu_{A_xO_y}^{\text{bulk}} = x\mu_A + y\mu_O \quad (6)$$

bounded by the elemental references:

$$\mu_A \leq \mu_A^{\text{elemental}}, \quad (7)$$

$$\mu_O \leq \mu_O^{\text{elemental}}, \quad (8)$$

and subject to inequalities that disfavour the formation of each competing phase A_iO_j :

$$i\mu_A + j\mu_O \leq \mu_{A_iO_j}^{\text{competing}}. \quad (9)$$

Construction of the chemical potential region requires consistent computational settings across all phases to extract reliable relative

energetics. In practice, however, achieving a single ‘one-size-fits-all’ description of every relevant phase is challenging: competing compounds can differ markedly from the host in their bonding, dimensionality and electronic structure. Layered phases may require van der Waals corrections, magnetic phases may need spin-polarized treatments even when the host is non-magnetic and heavy-element systems can be sensitive to SOC. When no single computational setup can simultaneously provide an accurate description of all phases, compromises are unavoidable, and the limitations of the chosen approach should be clearly stated. Correction schemes – such as the fitted elemental reference energies approach – can mitigate these inconsistencies⁸², but their applicability must be evaluated carefully; for example, neglect of SOC can introduce systematic errors (Box 1). Accurate representation of elemental reference states also remains essential: for instance, modelling O_2 in its triplet ground state rather than in a metastable singlet configuration avoids large systematic errors in the accessible range of μ_O .

Beyond quantitative errors, qualitative issues arise when relevant competing phases are omitted. Candidate competing phases are often identified using computational materials databases such as the Materials Project⁸³ or OQMD⁸⁴, followed by recomputation of their energies using the same computational settings as the defect calculations. In principle, as these databases report formation energies for a wide range of compounds, only the subset of phases that are bound to the host in chemical potential space needs to be included. However, confidently identifying these bounding phases is often non-trivial. Database coverage may be incomplete, and inaccuracies can lead to misidentification of the relevant tie-lines, especially in chemically complex or underexplored systems.

To mitigate these issues, practitioners may include additional candidate phases beyond those on the nominal convex hull. For example, the *doped* package⁸⁵ applies an uncertainty tolerance (for example, 0.05 eV per atom) to the formation energies reported in the Materials Project, treating them as approximate rather than exact. All compounds that fall within this stability threshold are included as potential competing phases, adopting a pragmatic strategy to account for uncertainty in formation energy data. This approach avoids over-reliance on a potentially incomplete or inaccurate phase diagram and allows for possible reordering of phase stabilities owing to known DFT limitations. Known gas-phase species (such as triplet O_2) are generated explicitly as isolated molecules. Additionally, it is often prudent to confirm that these databases are not missing experimentally resolved structures from resources such as the Inorganic Crystal Structure Database (ICSD)⁸⁶, especially in systems with known low-energy polymorphism or well-documented DFT failures.

Even so, databases are not exhaustive. In relatively unexplored composition spaces, critical phases may be missing altogether. In such cases, structure prediction methods may be required to identify previously unreported competing phases. For example, a structure search in the Li–Ni–O system revealed a new secondary phase that subtly shifted the stability window of the candidate lithium-ion cathode material $LiNiO_2$, with direct implications for its accessible chemical potentials and for the predicted defect chemistry^{7,87}. These challenges illustrate the dual risks of false negatives – omitting relevant phases that shift the stability region – and false positives – wasting resources on irrelevant or non-competing compounds. Careful assessment of phase diagram completeness and chemical potential assignments is essential for ensuring physically meaningful defect calculations.

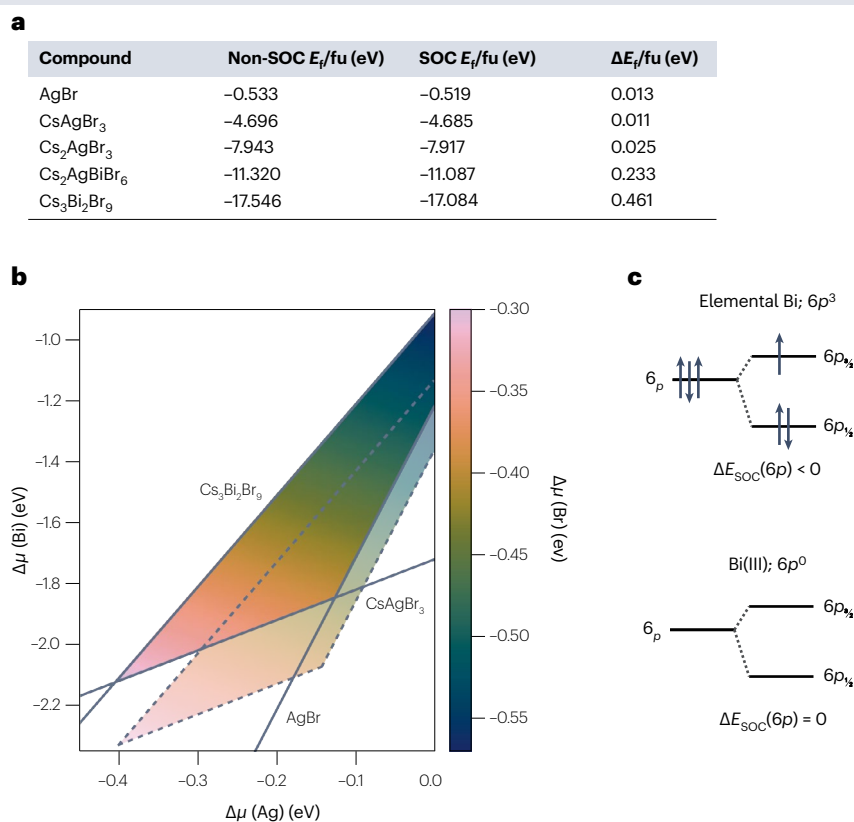
Box 1 | Role of spin–orbit coupling in thermodynamic stabilities and chemical potential limits

Often, spin–orbit coupling (SOC) is neglected in formation energy calculations for bulk compounds and, by extension, the chemical potentials used in defect studies. This simplification relies on the assumption that SOC introduces a rigid atomic energy shift that cancels out when calculating formation energies¹⁸⁰. However, SOC more accurately introduces a nearly rigid orbital energy shift, meaning that significant changes in orbital occupation between elemental reference states and their respective compounds can invalidate this assumption. This issue is particularly pronounced for elements where SOC strongly influences the valence electrons, making its impact oxidation-state-dependent.

To illustrate this, we calculate the formation energies of competing phases for the double perovskite $\text{Cs}_2\text{AgBiBr}_6$, which has been investigated for photovoltaic applications^{181–183}, using the PBEsol DFT functional and the automated analysis tools in *doped*, with and without SOC effects (see the figure, panel **a**).

Although the effect of SOC on the formation energies of Cs–Ag–Br compounds is minor, as expected, the effect on Bi-containing compounds is significant, with an ~ 0.23 eV energy shift per Bi atom. The difference in accessible chemical potentials is clear from the comparison between the chemical stability region obtained with SOC, shown with a solid outline in panel **b** of the figure, and that obtained excluding SOC, shown in faded colour with a dashed outline. SOC splits the energies of the valence $6p$ electrons, which are partially occupied in elemental Bi ($6p^3$) but unoccupied in Bi(III) compounds ($6p^0$, see the figure, panel **c**).

This results in oxidation-state-dependent energy shifts that significantly alter the chemical stability region, translating into



0.23 eV differences in defect formation energies at chemical potential limits, or a two orders of magnitude concentration difference for a synthesis temperature of 700 K. These discrepancies underscore the necessity of explicitly considering SOC effects when computing formation energies and chemical potentials for heavy-element systems.

Occasionally, defect formation energies evaluated under equilibrium conditions appear negative, by and large an unphysical prediction. When host instabilities have been excluded, such results typically reflect inconsistencies in the chemical-potential construction, most commonly missing competing phases, or μ -values located outside the true stability region of the host material. In some cases, these anomalies can even indicate the presence of previously unrecognized low-energy phases⁸⁸. Other common culprits for negative defect formation energies are inaccurate finite-size corrections, insufficiently large supercells or mismatching calculation parameters, issues that can be flagged by defect simulation packages⁸⁵. Negative formation energies in this context therefore point to issues in the computational setup, in the vast majority of cases, rather than indicating enthalpically favourable defect formation.

Relating computed chemical potentials to experimental synthesis conditions is a substantial challenge. Although X-rich/X-poor

limits provide a convenient starting point, establishing a more direct correspondence often requires thermochemical data or experimental calibration. In well-studied systems such as oxides, chemical potentials can sometimes be linked to oxygen partial pressures, enabling mapping between DFT-derived bounds and experimental growth conditions^{52,62,89,90}. Alternatively, first-principles models can predict the elemental chemical potential as a function of temperature and pressure, as has been performed for sulfur vapours^{91,92}, which can then be related to experimental conditions⁹³. Empirical techniques such as phase boundary mapping^{94–97} offer another route, identifying chemical potential limits by varying precursor ratios until secondary phases appear, providing direct experimental constraints on the stability region.

Finally, many technologically important materials are metastable or synthesized under non-equilibrium conditions. In these cases, the assumption of thermodynamic equilibrium used to define the chemical potential bounds may not hold. Growth techniques

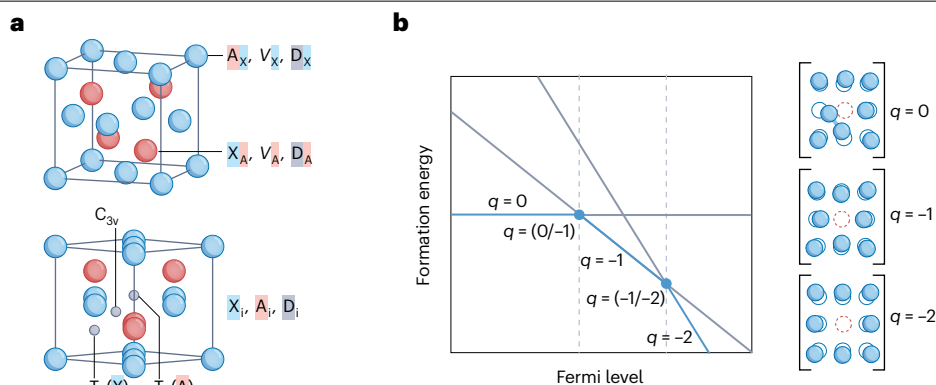


Fig. 2 | Enumerating defects and their charge states. **a**, Schematic of native and extrinsic point defect sites in an AX zincblende lattice, doped with species D. Red and blue spheres denote host atoms A and X, respectively. V_A and V_X are vacancies on the indicated sites; grey circles indicate potential interstitial positions; A_X and X_A are antisite defects; dopant substitutions appear as D_A or D_X . Interstitials (A_i , X_i and D_i) are located via Voronoi interpolation and classified by the local point symmetry (for example, T_d and C_{3v}) and coordination (for example, $T_d(A)$ and

$T_d(X)$). **b**, Schematic formation energy diagram of a defect in multiple charge states ($q=0, -1, -2$). Each line represents the formation energy of a specific charge state as a function of the Fermi level. The lowest-energy charge state at each Fermi level is highlighted, in which the stable defect configuration switches from one charge state to another. The schematics on the right show the local atomic structure for each charge state, illustrating the distinct local relaxations induced by changes in electron count.

such as molecular beam epitaxy or rapid quenching can access narrow chemical potential windows that do not appear in conventional phase diagrams^{98,99}. More accurate modelling may require pressure-dependent or temperature-dependent phase diagrams or incorporation of kinetic constraints¹⁰⁰. Because metastable conditions lack a well-defined thermodynamic reference, defect energetics in these systems carry additional uncertainty. Where possible, such studies should be cross-validated against experimental observations to ensure predictive reliability.

To support reproducibility, key details of the chemical potential construction should be reported. These include the computational settings used across host and competing phases; the treatment of elemental reference states; any corrections applied (such as for van der Waals interactions, magnetism or SOC); the source of phase stability data; and the selection criteria for competing phases. If uncertainty tolerances are applied to database energies or phase stabilities, their values and justification should be given. For metastable or non-equilibrium conditions, it is helpful to state how these were accounted for. Where comparisons to experiment are made – such as using oxygen partial pressures or phase boundary mapping – these should be described with sufficient context to relate them to the calculated chemical potentials.

Selecting defects and their charge states

Once the host structure is well defined and the chemical environment established, the next step in a defect study is to enumerate the relevant point defects and their possible charge states. Point defects are typically grouped into three categories: vacancies, substitutions (including antisites) and interstitials (Fig. 2a). Vacancies and substitutions are generally straightforward: removing an atom from the host forms a vacancy, replacing it with a different element from the host structure creates an antisite and replacing it with an extrinsic species gives a substitutional dopant defect. Only one calculation per symmetry-inequivalent site is required, as the site multiplicity factor N_x in equation (1) accounts for the degeneracy of equivalent sites.

Interstitial defects can be more challenging. Unlike vacancies and substitutions, in which the sites are defined by the host lattice,

interstitial sites must be located algorithmically. In simple crystals, interstitials may occupy high-symmetry positions, but in more complex structures, owing to low symmetry or complex bonding, more exploratory methods are required. Voronoi tessellation is a widely used method for identifying potential interstitial sites in crystals by partitioning space into regions associated with each atom, such that every point within a region is closer to its associated atom than to any other. The vertices where multiple Voronoi regions meet represent positions that are locally equidistant from several atoms and are often plausible candidates for interstitial positions. These candidate sites are typically further screened based on symmetry considerations, coordination environment and their distance to nearby host atoms^{85,101–103}. Alternative approaches include order-parameter-based topological analysis¹⁰⁴ and charge-density mapping, which identify trial sites near local maxima or minima in the electron density^{25,105,106} in the host material. Although the latter approach works well in ionic systems, it may miss relevant sites in covalent or metallic environments where charge distributions are more complex.

Chemical knowledge can further refine the defect set. Unlikely configurations – such as cation–anion antisites in strongly ionic materials or interstitials involving oversized ions – could be considered for omission to avoid unnecessary calculations. In addition, tightly bound atoms such as P^{5+} in PO_4^{3-} tetrahedra or W^{6+} in WO_6^{6-} octahedra can potentially be excluded from vacancy generation on the grounds of their high bond dissociation energies. Omissions should be justified.

False positives and false negatives are both concerns in defect enumeration. False positives arise when high-energy configurations are included unnecessarily, whereas false negatives occur when physically relevant defects are overlooked. These risks are particularly acute for interstitials, in which multiple trial sites may relax to the same configuration or worse, insufficient sampling may lead to important low-energy configurations being missed entirely. Transparent reporting of the generation strategy, filtering criteria and all tested configurations (even unstable ones) is essential for reproducibility and downstream reuse.

In addition, defect charge states must be selected for calculation. In semiconductors and insulators, each point defect may exhibit multiple

stable charge states q , which correspond to different local electron counts and geometries (Fig. 2b). These states behave as distinct chemical species with different formation energies and concentrations, and their thermodynamic populations are governed by Fermi-level-dependent Boltzmann statistics (equation (1)). Charge-state energetics are typically visualized through defect formation energy diagrams, in which transitions between the most stable charge states appear as discontinuities at the charge transition levels¹⁰⁷. We note that a common pitfall for new researchers in the field is the use of single-particle eigenvalues (densities of states) from defect supercell calculations for quantitative analyses. However, unlike delocalized bulk states, these localized states are affected by finite-size effects and are highly sensitive to local geometry (which changes as a function of their occupation, or charge state). As such, they can typically only be relied on for qualitative analyses (such as querying orbital characters and degree of localization).

A priori prediction of stable charge states is not straightforward, as stability depends on host and local electronic structure, local lattice flexibility and electrostatics. Although chemical guidance, through electron counting, can narrow the search space, it is not always reliable. Overly narrow selections risk false negatives, whereas broad searches increase computational cost and may yield irrelevant states (false positives). Approaches to mitigate these risks include screening charge states by oxidation-state plausibility^{102,105}, magnitude of host charges and energetic penalties associated with large deviations from nominal valence⁸⁵. Regardless of approach, all attempted charge states (whether stable or not) should be reported, and the selection methodology documented.

An illustrative example is the cation vacancy in ZnO (ref. 108). As is typical for cation vacancies, V_{Zn} is stable in the fully ionized negative charge state, V_{Zn}^{2-} , following the chemical intuition that removal of a Zn(II) ion yields a -2 relative charge, as well as in the spin-polarized singly negative and neutral charge states, V_{Zn}^{1-} and V_{Zn}^0 . However, in ZnO, the Zn vacancy is an exceptionally deep acceptor capable of strongly binding hole polarons on each of the four neighbouring oxygen ions. This results in the unexpected stabilization of the $+1$ and $+2$ charge states, which would be missed (false negatives) if only formal ionic charge states were considered. Similarly, nitrogen vacancies in III-nitrides have been found to stabilize in $+1$ and -1 charge states, contrary to a simple ionic picture¹⁰⁹. Another example of false-negative charge states is provided by the vacancies in Sb_2Se_3 . In addition to the expected fully ionized charge states, V_{Sb}^{3-} and V_{Se}^{2+} , both vacancies are amphoteric, with V_{Sb}^{1+} and V_{Se}^{2-} stabilized within the band gap¹¹⁰. Here, the rare four-electron negative- U behaviour arises from valence alternation and dimer/trimer formation identified by structural searches.

All stages of defect and charge-state enumeration and selection should be documented. This includes reporting the algorithms and thresholds used to identify interstitial sites, any omissions based on chemical or structural reasoning, the full set of tested configurations (including those found to be unstable or high in energy) and the selection criteria for trial charge states. Where charge states were excluded, the rationale – whether based on oxidation-state analysis, convergence behaviour or energetic infeasibility – should be provided. Explicitly reporting unstable or metastable configurations, even if filtered from subsequent analysis, enables transparent benchmarking and facilitates future comparisons.

Defect structure searching

Once defect and charge-state candidates are selected, the next step is to determine their relaxed geometries. The structural configuration of

a point defect strongly influences its properties. Historically, defects were modelled by making minimal modifications to the ideal bulk structure – for example, by removing, substituting or inserting atoms at lattice sites – followed by direct relaxation. Although straightforward, this approach often converges to local minima on the PES, missing lower-energy reconstructions^{36,43,111,112}.

Various strategies have been developed to address this challenge. Full structure-search methods, such as random sampling^{111,112} or genetic algorithms¹¹³, can uncover complex reconstructions but are computationally demanding. A more practical alternative is to apply targeted perturbations to the local environment, such as randomly displacing atoms around the defect site^{25,101}. These perturbations are often sufficient to break symmetry and help escape shallow basins in the PES. However, this approach is limited in its ability to identify the full range of possible defect reconstructions.

Recent advances have introduced chemically informed stochastic approaches such as the ShakeNBreak method^{43,49}, which incorporates information about excess charge and local bonding to generate diverse trial geometries. This method has been validated across a wide range of materials – including CdTe, GaAs, $\text{Sb}_2(\text{S/Se})_3$, CeO_2 , In_2O_3 , ZnO and anatase TiO_2 – and has successfully uncovered low-energy defect structures missed by conventional relaxations in diverse systems^{6,8,36,50,114}. However, there are cases in which this approach alone is insufficient or ill-suited, such as with ‘non-local’ reconstructions like split vacancies²⁸, or complex systems with many locally stable minima³, in which cases additional screening may be required.

In this context, it is important to recognize the limitations of the underlying DFT functional. Many reconstructions are driven by localized excess charge, which is impacted by self-interaction errors and over-delocalization. In particular, band gap underestimation and self-interaction can spuriously favour delocalized band-edge states over localized defect levels, leading to qualitatively incorrect defect geometries and charge distributions. As illustrated in Box 2, these shortcomings can result in failure to identify the true ground-state structure. Benchmark results using the ShakeNBreak method applied to known reconstructions in CdTe and Sb_2Se_3 show that semi-local functionals (GGA and meta-GGA) often fail to reproduce the correct energy ordering of defect geometries or even to locate the correct minimum at all. Although hybrid functionals such as HSE offer more accurate descriptions of the PES, their computational cost remains a limiting factor.

To balance accuracy and efficiency, it is common to perform the initial structure search using relaxed convergence criteria or lower-level methods, followed by refinement with higher-level calculations. Alternatively, surrogate models trained on high-quality defect data offer a promising route for fast and reliable PES exploration^{26,28}. These models can significantly reduce computational cost, but must be validated to ensure that they correctly reproduce the qualitative features of the PES.

Given the sensitivity of defect properties to structural relaxation, all studies should report the methods used for generating initial structures, including whether random displacements, chemically informed perturbations or structure-search algorithms were used. The level of theory and convergence criteria used for each stage of relaxation should also be documented, particularly when a multistep approach is used. It should be stated whether defect reconstructions were explicitly explored; if not, a justification – such as prior evidence of rigidity or computational constraints – should be provided. Where stochastic or machine-learned sampling was used, the number and diversity of trial structures should be reported, along with the criteria used to select final configurations.

Box 2 | Functional dependence of defect structure searching

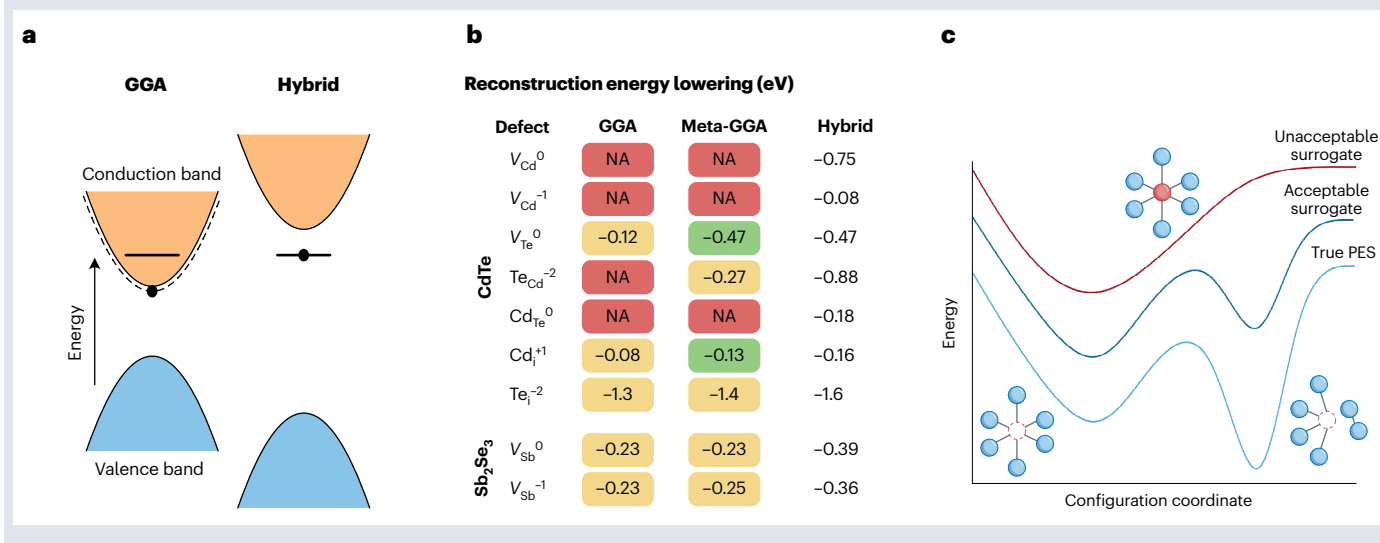
Defect reconstructions often arise owing to the presence of localized electronic states, which drive significant structural distortions. Semi-local density functional theory (DFT) functionals, such as GGA or meta-GGA, frequently fail to capture these effects owing to band gap underestimation and the delocalization of excess charge carriers. This is illustrated in panel **a** of the figure, in which a defect level that is properly localized within the gap under hybrid DFT is spuriously merged into the conduction band minimum when using GGA, destabilizing the true defect state.

Panel **b** of the figure presents benchmark results from structural searches for defects in CdTe and Sb₂Se₃ using the ShakeNBreak method. Each column compares the energy lowering achieved with GGA, meta-GGA and hybrid DFT. Green boxes indicate cases in which the reconstruction energy agreed within 0.05 eV of hybrid DFT; yellow boxes show cases in which the structure was found but with

a significant energy discrepancy; and red boxes mark cases in which the reconstruction was missed altogether. These results demonstrate that GGA and meta-GGA frequently fail to identify the correct defect geometry, even when subsequent hybrid DFT calculations are performed on the relaxed structures^{18,4}.

Panel **c** of the figure compares three representations of the defect potential energy surface (PES): the ‘true’ PES, typically approximated by hybrid or post-DFT methods; an acceptable surrogate, such as a low-convergence hybrid calculation or validated surrogate model^{26,28}; and an inadequate surrogate, which may entirely miss critical low-energy minima.

These findings underscore the fact that using low-level methods for structure search is not just quantitatively inaccurate but can also introduce fundamental qualitative errors. If relevant minima are not found, no post hoc correction can recover them.



Simulating the defective cell

The next step is to run the defect simulations – geometry relaxation and total-energy calculation – under converged settings needed for accurate energetics and electronic structure. If initial calculations were carried out with lower cost but acceptable surrogate settings – such as looser convergence criteria, reduced *k*-point sampling or a surrogate interatomic potential – as recommended with defect structure searching, for instance, then these calculations will continue from those starting points. Each defect/charge-state combination is simulated by adjusting the total number of electrons in the supercell to match the nominal charge state. Any net charge is compensated by a uniform background charge to maintain overall neutrality in the calculation, ensuring that the total energy remains finite under periodic conditions.

The choice of exchange-correlation functional has already been discussed in the context of defect structure searching, in which its impact on charge localization and structural PES was highlighted. In that stage, the goal was to recover the correct qualitative ground-state structure; here, the objective is to obtain quantitatively reliable formation energies and charge transition levels from the

final geometry. Although semi-local functionals (such as PBE) are computationally attractive – especially for large supercells – they suffer from well-known failures when applied to systems with localized electronic states. Semi-local DFT suffers from self-interaction errors and severe band gap underestimation, which destabilize localized defect levels relative to delocalized band-edge states, leading to inaccurate charge localization, incorrect structural reconstructions and ultimately unreliable defect energetics^{10,36,103,115–130}. Although a posteriori corrections such as band-edge alignment^{25,131} can partly mitigate band gap errors to correct transition levels, they fail to correct the associated charge-localization issues and do not influence the self-consistent energetics. Semi-local functionals should therefore be avoided whenever possible. If their use is unavoidable – for example, in high-throughput or screening workflows – the expected limitations and likely misrepresentations should be explicitly stated and, where possible, benchmarked against higher-level results.

Other simulation settings reflect both physical assumptions and practical constraints. Symmetry constraints, often applied in bulk calculations to reduce cost, must be disabled for defect systems to

allow for symmetry-breaking reconstructions. Lattice parameters should typically remain fixed at their bulk-optimized values, whereas internal atomic positions are relaxed to represent a dilute defect embedded in an extended host. Spin polarization is another critical consideration. Open-shell configurations frequently arise in defect systems owing to unpaired electrons, and spin-polarized calculations are necessary to correctly resolve their electronic structure and total energies. Although most odd-electron systems adopt doublet spin states ($S = 1/2$) and even-electron systems adopt singlets ($S = 0$), many exceptions exist, including bipolaronic and multipolaronic states (such as V_{Se}^0 in Se^6 , V_{Cd}^{0*} in CdTe^{116} or V_{P}^{1-} in NaP^{132}), molecular dimers (such as O_2 species in oxides⁷ or carbon pairs in silicon¹³³), multilocalized d/f -electron defects and orbital-degenerate configurations (such as the nitrogen-vacancy centre in diamond¹³⁴ or transition metal impurities in silicon¹³⁵). In these cases, trialling different spin initializations and magnetic configurations may be necessary to identify the correct ground state⁸⁷.

k -point sampling determines how reciprocal space is represented in the calculation. In principle, a sufficiently large defective supercell has a very small Brillouin zone, and a single Γ -point can approximate the reciprocal space integration well. Γ -point-only sampling is therefore often used for very large cells, particularly during structural relaxations. However, the required sampling depends on the physical size of the supercell and the dispersion of the relevant electronic states. Smaller supercells, or systems with shallow defect levels or light effective masses, often require denser meshes to capture the correct band curvature and defect–host interactions¹³⁶. For accurate formation energies, charge transition levels and electronic structure, k -point convergence testing remains essential, and as such checks should go beyond typical total-energy convergence. It is also important that the k -point mesh chosen for the primitive cell remains compatible with the intended supercell so that the reciprocal-space sampling density is preserved after replication. Choices in the primitive-cell mesh – such as how the grid subdivides when the cell is expanded or whether a shift is applied – can otherwise lead to unintended changes in the effective sampling. Ensuring consistent k -point sampling across unit cells and supercells is therefore non-trivial and should be reported clearly.

After the calculations are completed, outputs should be checked to ensure the results are physically reasonable: the relaxed geometry should make chemical sense, and partial charge, projected densities of states and spin densities can be used to assess whether the final electronic structure of the defect is physically reasonable. Discrepancies here can point to problems such as spurious delocalization, incorrect spin states or convergence to metastable geometries, and should be resolved before proceeding.

Addressing the limitations of the supercell method

A fundamental challenge in supercell-based defect calculations is the artificial periodicity imposed by boundary conditions. Simulation supercells introduce spurious interactions between periodic defect images (Fig. 3) and correspond to defect concentrations that are unrealistically large for most cases ($N_{\text{x}}^{-1} \sim 10^{-2}$ – 10^{-3} per host atom). These artefacts can arise from electrostatic interactions, wavefunction overlap, elastic distortions or magnetic coupling. An in-depth discussion of the influences of these interactions and others can be found in ref. 32.

Charged defects require particular care, because the long-range Coulomb interaction between periodic images decays slowly as q/er , often extending far beyond the supercell boundary. This weak decay means that charged systems retain significant spurious contributions

to energies even in large supercells, unless corrected. In addition to localized defect charges, finite supercell sizes introduce complications for shallow defects, where excess charge carriers occupy delocalized ‘perturbed host states’. This behaviour can manifest for neutral and charged defect states alike, with the carrier wavefunctions derived from host band-edge states and typically being weakly bound to the defect. Similar to Wannier–Mott excitons, these hydrogenic states have small binding energies (1–100 meV) and large spatial extent. Although their long-range electrostatic effects are diminished, the delocalization of these carriers across many unit cells results in overlap with their periodic images. If the defect site is charged, spurious excitonic interactions with the confined carrier wavefunctions will also occur. Additionally, shallow states can result in ‘band filling’: the occupation of band-edge-derived states with unphysically large populations. Here, the high carrier density in the finite supercell corresponds to the occupation of a substantial energy range within the host conduction or valence bands, whereas in reality only the low-energy edge is occupied (outside degenerate doping). Combined with the wavefunction confinement of the shallow state, this occupation of the band edges can also lead to spurious hybridization between the defect and bulk states that depends sensitively on the supercell size and shape.

The first step towards mitigating these effects is careful supercell design. Optimal supercells balance the need to minimize image interactions against computational cost. Various algorithms have been developed to generate optimal simulation supercells, such as expanding the cell to encompass a target cube size^{105,106}, finding the supercell that best matches a target shape^{137,138} or directly minimizing the number of atoms for a given minimum image distance⁸⁵. Defect separation strongly influences the predictability of electrostatic scaling and the accuracy of subsequent corrections. A common heuristic is a minimum of 10 Å between periodic images, which typically limits formation energy errors to below 0.1 eV when combined with appropriate corrections⁷⁸. However, this threshold is not universally sufficient: materials with strong anisotropies or weak dielectric screening may require larger separations^{77,103,139}. In low-dimensional or layered materials, weighting image separation by the dielectric tensor or k -point mesh can be more effective. Additionally, commensurability with structural motifs (such as octahedral tilts in perovskites) can help avoid artificial strain (as discussed in the section about the host material).

Even in large supercells, long-range electrostatic interactions with charged defects necessitate explicit correction schemes. Early approaches, such as the Makov–Payne correction and its extensions^{72,140–142}, treat the defect as a point charge screened by the bulk dielectric. More accurate methods, such as the Freysoldt–Neugebauer–Van de Walle (FNV) scheme⁷⁷ and its anisotropic extension (eFNV)⁷⁸, explicitly model the defect potential and better account for the dielectric environment, whereas more recent work has looked to incorporate these corrections self-consistently within DFT calculations^{143–145}. Further extensions have been introduced for charged defects at surfaces, interfaces and 2D materials^{139,146}, optical transitions at defects^{79,147} and polarization charges in configuration coordinate diagrams¹⁴⁸.

For defects with excess carriers, the finite-size errors from band filling and wavefunction overlap require different treatment. Band-filling corrections¹⁰¹ can remove the artificial total-energy contribution from partially filled bands, but cannot address spurious hybridization, overlap or excitonic effects. Moreover, excitonic and band-filling effects often act in opposite directions, producing

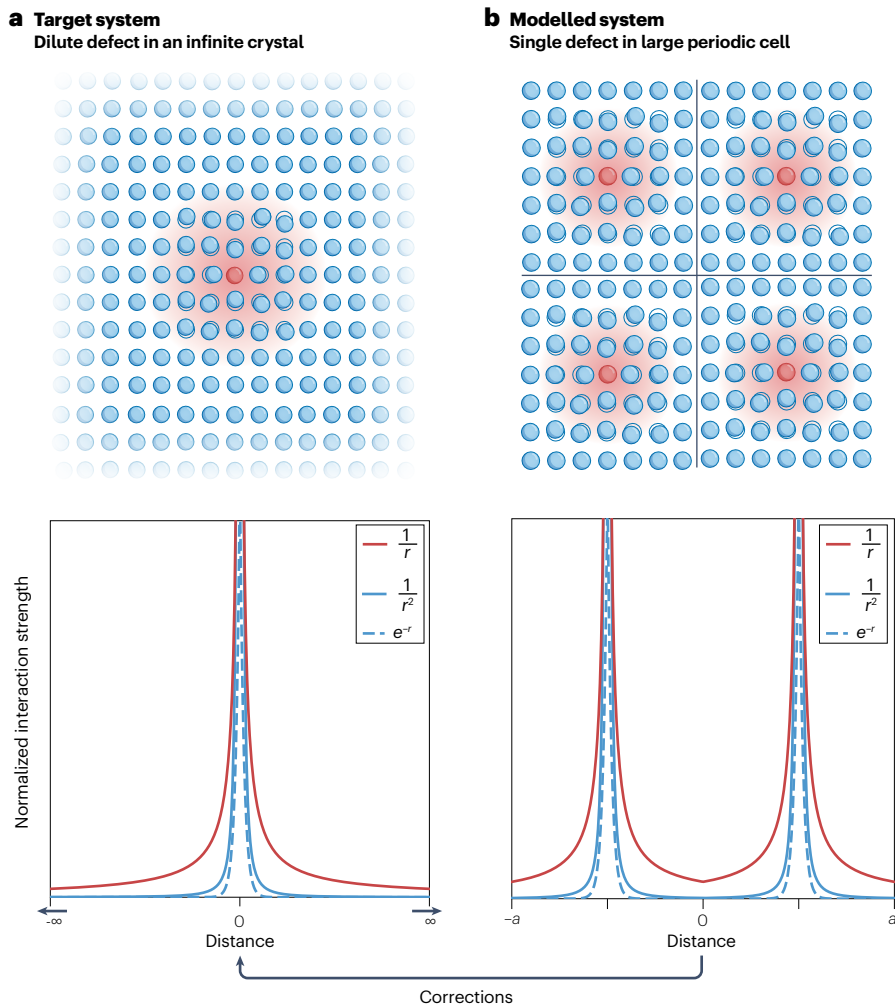


Fig. 3 | Defect modelling in periodic supercells.

a, Idealized target system: a single, isolated defect embedded in an infinite crystal. The defect perturbation (such as charge density distortion, strain or wavefunction overlap) decays smoothly with distance from the defect site, r , as seen in the lower panel, showing the schematic decay of different defect-induced perturbations: electrostatics $\propto \frac{1}{r}$, displacement field $\propto \frac{1}{r^2}$ and localized wavefunction overlap $\propto e^{-r}$. **b**, Practical model system: a single defect in a finite periodic supercell. Periodic boundary conditions lead to spurious interactions between defect images, particularly for long-range perturbations. As seen in the lower panel, long-range interactions persist across the cell and must be corrected to recover the dilute-limit behaviour. Corrections aim to remove these artefacts and approximate the isolated defect limit.

deceptive error cancellation. Ultimately, precise calculation of shallow defect binding energies at present necessitates extrapolation to very large cell sizes. That said, effective-mass models ($E_b = 13.6 \text{ eV} \times \bar{m}/\epsilon^2$), which are independent of the defect identity, often predict shallow binding energies with good accuracy, particularly for isotropic and dispersive systems¹¹⁹. We note that automated orbital and eigenvalue analysis tools such as those in *doped*⁸⁵ and *pydefect*²⁵ can aid the identification of perturbed host states.

An alternative to explicit correction schemes is supercell extrapolation, in which defect formation energies are computed for a sequence of increasingly large cells and extrapolated to the infinite-size limit¹⁴⁹. This approach makes fewer assumptions about charge localization or dielectric screening and can reveal how different finite-size effects scale, but is computationally demanding, particularly for shallow states spanning thousands of atoms¹¹⁹. Recent work on the scaling behaviour of hybrid DFT¹⁵⁰ may improve the feasibility of this approach.

Ultimately, no single strategy works for all cases. Accurate defect energetics require judicious choices based on the material, dimensionality and defect type. Here, reproducibility requires explicit reporting of the supercell size and shape, the minimum image distance and the justification for their selection; the correction scheme used (such

as Makov–Payne, FNV and eFNV) and any assumptions about charge localization; the criteria for identifying shallow states; and any treatment of band-filling or excitonic effects, including any anticipated error cancellation. For extrapolation strategies, the supercell series, scaling form and fit quality should be documented to allow reproducible comparison across studies.

Quantifying defect populations

Under (quasi)equilibrium conditions, defect formation energies determine the observable defect concentrations. From equation (1) and the approximation of free energies with internal energies, the equilibrium concentration $[X^q]$ of a dilute defect X in charge state q is given by^{36,151}:

$$[X^q] = N_X g_{X^q} \exp \left(\frac{-E_f^{X^q}(E_F, \mu)}{k_B T} \right), \quad (10)$$

in which N_X is the number of sites at which X can form, g_{X^q} is the defect degeneracy factor, k_B is Boltzmann's constant, T is the temperature and $E_f^{X^q}(E_F, \mu)$ is the formation energy, which depends on the Fermi level E_F and the set of chemical potentials $\{\mu_i\}$. At high defect

concentrations, the competition for available lattice sites modifies the expression to a Fermi–Dirac-like form^{152,153}:

$$[X^q] = N_X \frac{g_{X^q} \exp\left(-\frac{E_f^{X^q}(E_F, \mu)}{k_B T}\right)}{1 + \sum_{X'} \sum_{q'} g_{X'q'} \exp\left(-\frac{E_f^{X'q'}(E_F, \mu)}{k_B T}\right)}, \quad (11)$$

in which the denominator accounts for all defects X' and charge states q' competing for the same type of site. Configurational degeneracies $g_{X'q'}$ can significantly affect site competition^{52,85}, especially when differing by orders of magnitude. However, when site competition is important, defect–defect interactions often become important as well and may require beyond-dilute-limit treatments for quantitative accuracy.

In most materials, equilibrium defect concentrations at room temperature are rarely realized owing to high kinetic barriers. Instead, it is often assumed that defects form under high-temperature synthesis or annealing conditions and become kinetically frozen during cooling^{154–156}. In this picture, defect concentrations are calculated at a high processing temperature $T_{\text{processing}}$ and then fixed while charge states and free carriers re-equilibrate at a lower operating temperature $T_{\text{operating}}$. This constrained equilibrium model – in which total defect concentrations are fixed but their charge-state distribution varies with E_F – forms the basis of many defect chemistry calculations and is outlined in Fig. 4.

This framework, however, can fail in systems in which defects remain mobile at or near room temperature. Such behaviour is well documented in ionic conductors^{12,87,157–159}, halide perovskites^{160,161} and semiconductors with mobile interstitials^{55,162–164}. In these systems, redistribution or annihilation of mobile defects alters the

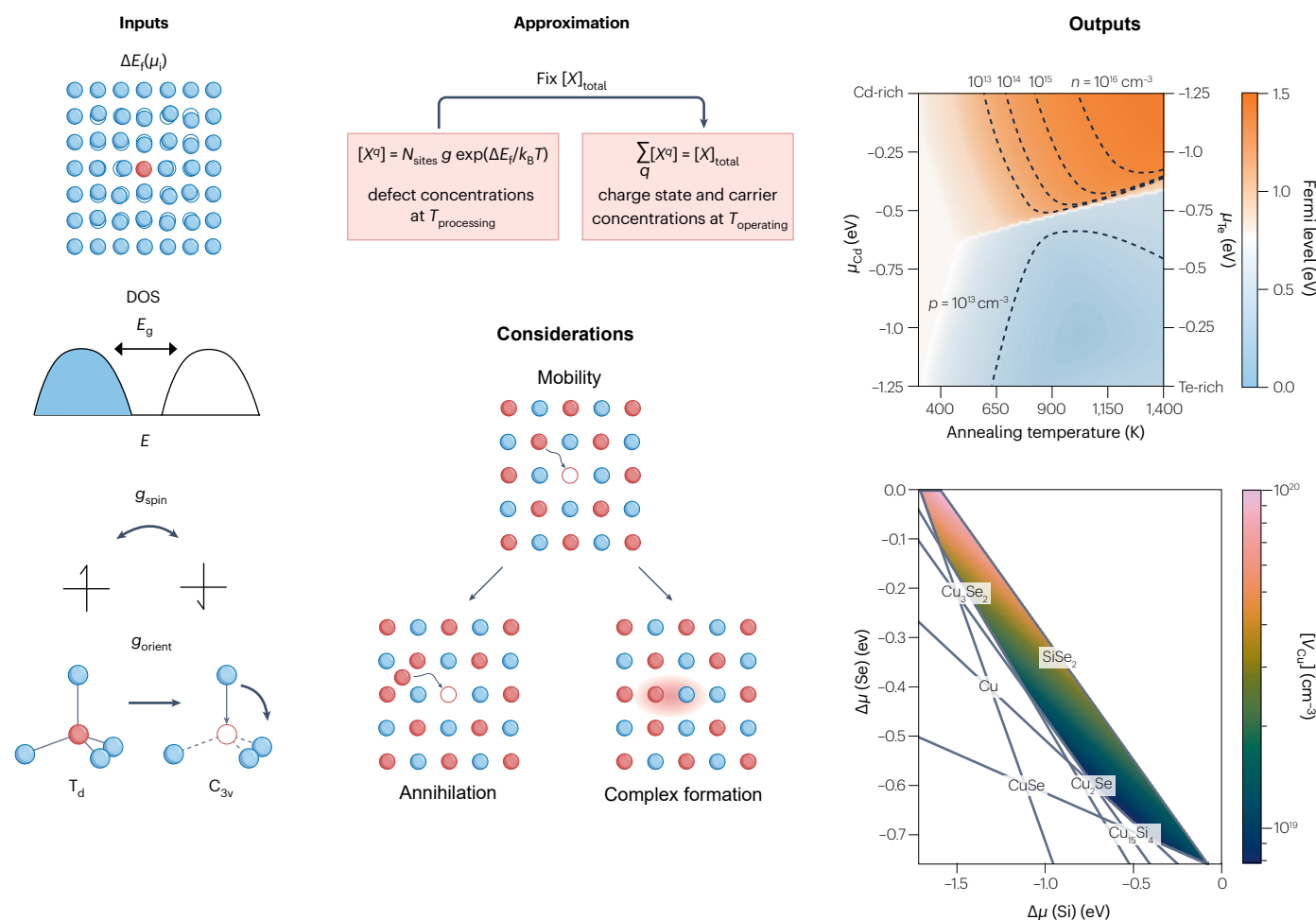


Fig. 4 | Workflow for determining defect and carrier concentrations under quasiequilibrium conditions. Inputs include accurate defect formation energies as a function of chemical potential $\Delta E_f(X^q)$, the electronic density of states (DOS) and configurational degeneracies (such as spin and orientational). Defect concentrations are computed at the processing temperature $T_{\text{processing}}$ in which thermal equilibrium is assumed. These concentrations are then frozen in and held fixed during cooling. At the lower operating temperature $T_{\text{operating}}$, only charge-state and carrier populations are re-equilibrated by solving

the charge-neutrality condition. Additional considerations, such as defect mobility, annihilation and complex formation, may invalidate the frozen-defect approximation and necessitate kinetic modelling. Example outputs include the dependence of the Fermi level in CdTe on the chemical potential and on the annealing temperature (top right; the dashed lines show the corresponding carrier concentrations)⁶² and the concentration of copper vacancies as a function of chemical potentials in Cu_2SiSe_3 (ref. 2) (bottom right).

overall population, violating the frozen-in assumption. Similarly, non-equilibrium growth methods (such as molecular-beam epitaxy and solution processing) may never reach thermodynamic equilibrium, further invalidating this approximation. In rare cases, charge-state re-equilibration may also be incomplete owing to slow carrier capture or emission. More realistic defect population models for these cases require explicit kinetic treatment^{85,164–166}, although such approaches remain computationally demanding, even with machine-learned interatomic potentials.

A self-consistent Fermi level E_F must be determined to ensure charge neutrality between all charged species in the system. This is imposed via the condition:

$$0 = \sum_{X^q} q[X^q] + p_0 - n_0, \quad (12)$$

in which n_0 and p_0 are the electron and hole concentrations, respectively, calculated from the electronic density of states (DOS) $D(E)$ using Fermi–Dirac statistics:

$$\begin{aligned} n_0 &= \int_{E_{\text{CBM}}}^{\infty} \frac{D(E)}{e^{(E-E_F)/k_B T} + 1} dE, \quad p_0 \\ &= \int_{-\infty}^{E_{\text{VBM}}} \left(1 - \frac{1}{e^{(E-E_F)/k_B T} + 1} \right) D(E) dE. \end{aligned} \quad (13)$$

As $[X^q]$, n_0 and p_0 all depend on E_F , these quantities must be solved self-consistently^{156,165}. Omission of significant charged defects from this process – especially low-energy species – can result in large errors in both predicted carrier concentrations and defect populations, underlining the importance of minimizing false negatives during defect enumeration.

A key variable in these expressions is the degeneracy factor g_{X^q} , which reflects the number of equivalent configurations a defect can adopt^{25,36,167}. For instance, spin degeneracy occurs when a defect introduces unpaired electrons, creating multiple equivalent spin states – which can be particularly important and complex in magnetic systems. Orientational or geometric degeneracy arises when a defect can adopt multiple equivalent spatial orientations, occurring when the defect reduces the symmetry of the host site – such as multiple equivalent Jahn–Teller-like distortions of a vacant octahedron. Electronic and vibrational degrees of freedom can also contribute to the defect-induced entropy change and thus degeneracy pre-factor³⁶ or can be incorporated within the formation energy term. Defect concentrations are directly proportional to the degeneracy factor; $[X^q] \propto g_{X^q}$ (equation (11)), and g_{X^q} can reach values up to ~100 (ref. 167). As such, failure to account for these forms of degeneracy can introduce significant errors in predicted defect concentrations, highlighting the need for a careful and comprehensive treatment in defect studies^{36,167}. Modern tools now support automated degeneracy assignment⁸⁵, improving reliability and reproducibility.

The accuracy of the calculated DOS, particularly near the conduction and valence band edges, is also crucial. Poor k -point sampling or inappropriate interpolation can significantly distort the band-edge DOS, leading to erroneous carrier and defect concentrations¹⁶⁸. This issue can be particularly pronounced for materials with complex band structures, particularly if band extrema are located away from high-symmetry points, as for Si¹⁶⁹, chiral compounds^{6,132,170} and many thermoelectric materials^{136,171}. Tetrahedron interpolation typically offers more reliable resolution near critical points. All choices regarding DOS construction – k -point density, smearing method and convergence thresholds – should be explicitly reported.

Finally, temperature-dependent effects beyond configurational entropy – including vibrational entropy, thermal expansion and electron–phonon coupling – can substantially modify defect energetics and Fermi-level positions, particularly for materials processed under high temperatures^{36,55,166}. Although vibrational free energies are costly to compute, machine-learned interatomic potentials offer promising speed-ups^{28,55,56}. Beyond defect formation energies, thermal renormalization of band edges can drastically alter carrier concentrations and thus defect populations and Fermi level (shifting E_F by >1 eV in extreme cases), while being much more computationally tractable^{52,62,85,166}. Temperature dependence of competing phases, particularly for gases such as O₂, F₂ or sulfur vapour, can substantially alter the chemical potential landscape^{50,52,85,113,153,172,173}. These effects are increasingly tractable and should be included where feasible.

All defect concentration calculations should report the processing and operating temperatures used, the assumed equilibrium, frozen-in or other approximations and whether kinetic effects were considered. The method for calculating the Fermi level, including the electronic DOS, k -point mesh, smearing method and integration scheme, should be specified. Degeneracy factors should be stated, especially in systems with multiple symmetry-equivalent configurations or spin states, and any automated tools used for assignment should be cited. If site competition or non-dilute effects are included, the competing species and their degeneracies should also be enumerated. Additionally, any treatment of temperature-dependent contributions to defect formation energies or band edges should be specified.

Practical steps to reproducible defect analysis

We have discussed key considerations for executing accurate defect supercell calculations. Given the many steps and choices to be made in the defect simulation workflow (such as DFT functional, bulk lattice parameters, charge-state ranges, structure searching, supercell size, finite-size corrections, kinetic effects and so on), there are multiple potential sources of discrepancies in final predictions. This issue highlights the need for established standards in the reporting of methodologies and sharing of data from defect investigations. In addition to boosting transparency and credibility, comprehensive reporting and data provision will allow us to build upon previous studies faster and more reliably, to improve our understanding of the impacts of various approximations and simulation strategies and to instil greater confidence in our results. When certain approximations are necessary, owing to computational overheads, for instance, these practices would help to indicate which property predictions are more or less prone to be affected. Overall, the improved reliability, interpretability and interoperability afforded by transparent, standardized reporting and data-sharing practices will reduce redundancy and boost the overall effectiveness of theoretical defect research.

The goal of establishing a comprehensive reporting and data-sharing practice for defect studies is to enable researchers to readily build on prior work. The considerations discussed at the end of each preceding section outline the minimum reporting standards we consider essential for best-practice defect studies. In Table 2, we list this combined set of minimum details, which we believe should be reported in all computational defect studies, along with open-access sharing of the underlying data. These details should allow any competent defect theorist to reliably reproduce and build on all predictions from a given study, and we should demand this minimum standard of ourselves as a community.

The recent development of software packages to manage the defect simulation workflow (Table 1) substantially reduces the human

Table 2 | Checklist of information required for reproducible point-defect simulations in crystals

Aspect	Information to report
1. Setup	
Unit cell	Lattice constants, space group
Stability	Thermodynamic and dynamic stabilities, any treatment of metastability
Supercell	Expansion matrix and image distance
Defect catalogue	Site enumeration algorithm; rationale for any omitted sites
Charge states tested	List with selection rules for charge-state ranges
2. Calculations	
Energy calculator	Code name/version; DFT functional or otherwise; SOC and/or vdW inclusion
Calculation settings	Basis set specification, <i>k</i> -mesh, electronic convergence criteria, pseudopotential choices
Geometry optimization	Relaxation protocol, including structure-searching, and trial and relaxed structures
Dielectric tensor	Value and provenance (computed or experimental)
Chemical potentials	Phase set, reference energies, SOC and/or vdW choices
3. Post-processing	
Finite-size corrections	Scheme(s) used and per-defect values
Thermodynamic analysis	Fermi-level solver; carrier and defect statistics; temperature; degeneracy factors
Analysis software	Code name and version, notebooks and scripts; package metadata files
Raw data archive	Full input and output files in an open repository

DFT, density functional theory; SOC, spin-orbit coupling; vdW, van der Waals.

effort of collating, reporting and sharing these data. In many cases, these tools can directly export parsed data and associated information to lightweight metadata files, which can then be provided along with other relevant data in open-access repositories. For instance, in the doped Python library⁸⁵, each data class can be exported to file with one line of code (for example, `to_csv()` or `to_json()`), saving all attached information (defect placement, charge-state ranges, host structure, symmetries and degeneracies, finite-size corrections and so on) and allowing these data to be reloaded and inspected with equal ease. Many of these codes can be used to generate summary tables of key quantities such as formation energy contributions, charge transition levels, point symmetries, degeneracy factors, chemical potential limits and more, which can be pasted into supporting information files for enhanced queryability^{6,52,62}. Additionally, some defect simulation packages^{25,85} use built-in sanity checks to identify common errors, such as inconsistent calculation parameters, mismatched bulk/defect supercells or chemical potentials or spurious charge delocalization. Reporting should include any custom workflows or defect analysis packages used. If new tools were developed, making them available to the broader community will substantially enhance reproducibility

and enable other researchers to adapt or extend the methods. All code versions should be cited to ensure consistency and traceability, ideally accompanied by a tabulation of key parameters and corrections to facilitate accessibility and interoperability.

To advance the field of defect simulation, transparency in methodology and data is absolutely essential. The baseline expectation should be to share complete data sets of inputs and results, in alignment with the FAIR data principles³⁷, rather than merely promising access on ‘reasonable request’, a practice that risks data loss over time and limits accessibility. Sharing the full input and output sets, where permissible, not only ensures reproducibility and independent validation of findings but also creates a robust foundation for future research. In addition, detailed reporting of calculation parameters, structural definitions, workflow choices, computational toolkits and correction schemes ensures that future research builds upon a reliable computational foundation. As a community, we should demand that these minimum standards of reporting and open-access data provision are adhered to, for the benefit of all. Along with allowing greater comparability between studies, this transparency will boost the credibility of computational defect research and promote the evolution of the field, helping to push the frontiers of defect research while fostering innovation and rigour. Establishing such solid methodological foundations also provides the strongest basis for extending defect studies with confidence to more complex material classes, such as amorphous and disordered systems^{174–176}. In the absence of a single crystalline environment, defects must be understood across an ensemble of local atomic configurations, with energetics, charge states and electronic responses drawn from a distribution rather than a single value. Alloys, amorphous phases and natively disordered compounds therefore call for ensemble-based approaches and, in many cases, a substantial change in how defects are generated, relaxed, sampled and interpreted. The principles we outline here – transparent methods, careful convergence, consistent thermodynamic referencing and clear reporting – form a natural platform on which future ensemble-driven defect methodologies can be built.

Published online: 19 January 2026

References

1. Catlow, C., Corish, J., Jacobs, P. & Lidiard, A. The thermodynamics of characteristic defect parameters. *J. Phys. C Solid State Phys.* **14**, L121 (1981).
2. Nicolson, A., Kavanagh, S. R., Savory, C. N., Watson, G. W. & Scanlon, D. O. Cu₂SiSe₃ as a promising solar absorber: harnessing cation dissimilarity to avoid killer antisites. *J. Mater. Chem. A* **11**, 14833–14839 (2023).
3. Wang, X., Kavanagh, S. R., Scanlon, D. O. & Walsh, A. Upper efficiency limit of Sb₂Se₃ solar cells. *Joule* **8**, 2105–2122 (2024).
4. Yuan, Z. et al. Discovery of the zintl-phosphide BaCd₂P₂ as a long carrier lifetime and stable solar absorber. *Joule* **8**, 1412–1429 (2024).
5. Whalley, L. D. et al. Giant Huang–Rhys factor for electron capture by the iodine interstitial in perovskite solar cells. *J. Am. Chem. Soc.* **143**, 9123–9128 (2021).
6. Kavanagh, S. R. et al. Intrinsic point defect tolerance in selenium for indoor and tandem photovoltaics. *Energy Environ. Sci.* **18**, 4431–4446 (2025).
7. Squires, A. G., Ganeshkumar, L., Savory, C. N., Kavanagh, S. R. & Scanlon, D. O. Oxygen dimerization as a defect-driven process in bulk LiNiO₂. *ACS Energy Letters* <https://doi.org/10.1021/acsenergylett.4c01307> (2024).
8. Murdock, B. E. et al. Li-site defects induce formation of Li-rich impurity phases: implications for charge distribution and performance of LiNi_{0.5-x}M_xMn_{1.5}O₄ cathodes (M = Fe and Mg; x = 0.05–0.2). *Adv. Mater.* <https://doi.org/10.1002/adma.202400343> (2024).
9. Gorai, P., Famprakis, T., Singh, B., Stevanović, V. & Canepa, P. Devil is in the defects: electronic conductivity in solid electrolytes. *Chem. Mater.* **33**, 7484–7498 (2021).
10. Hoang, K. & Johannes, M. D. Defect chemistry in layered transition-metal oxides from screened hybrid density functional calculations. *J. Mater. Chem. A* **2**, 5224–5235 (2014).
11. Eggstad, K., Selbach, S. M. & Williamson, B. A. D. Doping implications of Li solid state electrolyte Li_{1-x}La_{2-x}Zr₂O₁₂. *J. Mater. Chem. A* **12**, 15666–15675 (2024).
12. Squires, A. G. et al. Low electronic conductivity of Li_{1-x}La_{2-x}Zr₂O₁₂ solid electrolytes from first principles. *Phys. Rev. Mater.* <https://doi.org/10.1103/PhysRevMaterials.6.085401> (2022).

13. Hu, J. et al. Enabling ionic transport in Li_2AlP_2 : the roles of defects and disorder. *J. Mater. Chem. A* **13**, 6427–6439 (2025).
14. Weber, J. R. et al. Quantum computing with defects. *Proc. Natl. Acad. Sci. USA* **107**, 8513–8518 (2010).
15. Turiansky, M. E. & Van de Walle, C. G. Dielectric loss due to charged-defect acoustic phonon emission. *APL Quantum* <https://doi.org/10.1063/5.0205532> (2024).
16. Jackson, A. J. et al. Computational prediction and experimental realization of earth-abundant transparent conducting oxide Ga-doped ZnSb_2O_6 . *ACS Energy Lett.* **7**, 3807–3816 (2022).
17. Willis, J. et al. Limits to hole mobility and doping in copper iodide. *Chem. Mater.* **35**, 8995–9006 (2023).
18. Li, K., Willis, J., Kavanagh, S. R. & Scanlon, D. O. Computational prediction of an antimony-based *n*-type transparent conducting oxide: F-doped Sb_2O_6 . *Chem. Mater.* **36**, 2907–2916 (2024).
19. Varley, J. B., Peelaers, H., Janotti, A. & Van De Walle, C. G. Hydrogenated cation vacancies in semiconducting oxides. *J. Phys. Condens. Matter* **23**, 334212 (2011).
20. Sabino, F. P., Chatratin, I., Janotti, A. & Dalpian, G. M. Hole conductivity through a defect band in ZnGa_2O_6 . *Phys. Rev. Mater.* <https://doi.org/10.1103/PhysRevMaterials.6.064602> (2022).
21. Hachmione, S., Ganose, A. M., Sullivan, M. B. & Scanlon, D. O. Exploring the thermoelectric potential of MgB_2 : electronic band structure, transport properties, and defect chemistry. *Chem. Mater.* **36**, 6062–6073 (2024).
22. Xiong, Y. et al. High-throughput identification of spin-photon interfaces in silicon. *Sci. Adv.* <https://doi.org/10.1126/sciadv.adh8617> (2023).
23. Yan, Q., Kar, S., Chowdhury, S. & Bansil, A. The case for a defect genome initiative. *Adv. Mater.* <https://doi.org/10.1002/adma.202303098> (2024).
24. Shen, J.-X., Voss, L. F. & Varley, J. B. Simulating charged defects at database scale. *J. Appl. Phys.* <https://doi.org/10.1063/5.0203124> (2024).
25. Kumagai, Y., Tsunoda, N., Takahashi, A. & Oba, F. Insights into oxygen vacancies from high-throughput first-principles calculations. *Phys. Rev. Mater.* **5**, 123803 (2021).
26. Mosquera-Lois, I., Kavanagh, S. R., Ganose, A. M. & Walsh, A. Machine-learning structural reconstructions for accelerated point defect calculations. *npj Comput. Mater.* **10**, 121 (2024).
27. Mannodi-Kanakkithodi, A. et al. Machine-learned impurity level prediction for semiconductors: the example of Cd-based chalcogenides. *npj Comput. Mater.* <https://doi.org/10.1038/s41524-020-0296-7> (2020).
28. Kavanagh, S. R. Identifying split vacancy defects with machine-learned foundation models and electrostatics. *J. Phys. Energy* **7**, 045002 (2025).
29. Coletti, G. et al. Impact of metal contamination in silicon solar cells. *Adv. Funct. Mater.* **21**, 879–890 (2011).
30. Luo, T. et al. Creation of nitrogen-vacancy centers in chemical vapor deposition diamond for sensing applications. *New J. Phys.* **24**, 033030 (2022).
31. Mosquera-Lois, I., Kavanagh, S. R., Klarbring, J., Tolborg, K. & Walsh, A. Imperfections are not 0 K: free energy of point defects in crystals. *Chem. Soc. Rev.* **52**, 5812–5826 (2023).
32. Freysoldt, C. et al. First-principles calculations for point defects in solids. *Rev. Mod. Phys.* **86**, 253–305 (2014).
33. Kim, S., Hood, S. N., Park, J.-S., Whalley, L. D. & Walsh, A. Quick-start guide for first-principles modelling of point defects in crystalline materials. *J. Phys. Energy* **2**, 036001 (2020).
34. Kröger, F. A. & Vink, H. J. in *Solid State Physics* (eds Seitz, F. & Turnbull, D.) Vol. 3, 307–435 (Academic Press, 1956).
35. Zhang, S. & Northrup, J. Chemical potential dependence of defect formation energies in GaAs: application to Ga self-diffusion. *Phys. Rev. Lett.* **67**, 2339–2342 (1991).
36. Mosquera-Lois, I., Kavanagh, S. R., Walsh, A. & Scanlon, D. O. Identifying the ground state structures of point defects in solids. *npj Comput. Mater.* <https://doi.org/10.1038/s41524-023-00973-1> (2023).
37. Wilkinson, M. D. et al. The FAIR Guiding Principles for scientific data management and stewardship. *Sci. Data* **3**, 160018 (2016).
38. Drabold, D. A. & Estreicher, S. (eds) *Theory of Defects in Semiconductors* 2007 edn (Springer, 2006).
39. Lidiard, A. B. The Mott–Littleton method: an introductory survey. *J. Chem. Soc. Faraday Trans. 2* **85**, 341–349 (1989).
40. Zhu, L. et al. Formation of intrinsic point defects in AlN: a study of donor and acceptor characteristics using hybrid QM/MM techniques. *J. Mater. Chem. A* **12**, 25449–25464 (2024).
41. Zhang, X. et al. Toward a consistent prediction of defect chemistry in CeO_2 . *Chem. Mater.* **35**, 207–227 (2023).
42. De Souza, R. & Harrington, G. Revisiting point defects in ionic solids and semiconductors. *Nat. Mater.* **22**, 794–797 (2023).
43. Mosquera-Lois, I. & Kavanagh, S. R. In search of hidden defects. *Matter* **4**, 2602–2605 (2021).
44. Gao, W. & Tkatchenko, A. Electronic structure and van der Waals interactions in the stability and mobility of point defects in semiconductors. *Phys. Rev. Lett.* <https://doi.org/10.1103/PhysRevLett.111.045501> (2013).
45. Zhang, X. & Wei, S.-H. Origin of efficiency enhancement by lattice expansion in hybrid-perovskite solar cells. *Phys. Rev. Lett.* **128**, 136401 (2022).
46. Meesala, S. et al. Strain engineering of the silicon-vacancy center in diamond. *Phys. Rev. B* <https://doi.org/10.1103/PhysRevB.97.205444> (2018).
47. Santra, P. et al. Strain-modulated defect engineering of two-dimensional materials. *npj 2D Mater. Appl.* <https://doi.org/10.1038/s41699-024-00472-x> (2024).
48. Gouveia, J. D. & Coutinho, J. Can we rely on hybrid-DFT energies of solid-state problems with local-DFT geometries? *Electron. Struct.* **1**, 015008 (2019).
49. Mosquera-Lois, I., Kavanagh, S. R., Walsh, A. & Scanlon, D. O. ShakeNBreak: navigating the defect configurational landscape. *J. Open Source Softw.* **7**, 4817 (2022).
50. Neilson, W. D., Rizk, J., Cooper, M. W. D. & Andersson, D. A. Oxygen potential, uranium diffusion, and defect chemistry in UO_{2-x} : a density functional theory study. *J. Phys. Chem. C* **128**, 21559–21571 (2024).
51. Zhao, X.-G., Dalpian, G. M., Wang, Z. & Zunger, A. Polymorphous nature of cubic halide perovskites. *Phys. Rev. B* <https://doi.org/10.1103/PhysRevB.101.155137> (2020).
52. Ogawa, K., Kavanagh, S. R., Oba, F. & Walsh, A. Defect tolerance via external passivation in the photocatalyst SrTiO_3 :Al. *J. Am. Chem. Soc.* **147**, 23180–23191 (2025).
53. Kayastha, P., Fransson, E., Erhart, P. & Whalley, L. Octahedral tilt-driven phase transitions in BaZrS_3 chalcogenide perovskite. *J. Phys. Chem. Lett.* **16**, 2064–2071 (2025).
54. Choi, M., Oba, F., Kumagai, Y. & Tanaka, I. Anti-ferrodistortive-like oxygen-octahedron rotation induced by the oxygen vacancy in cubic SrTiO_3 . *Adv. Mater.* **25**, 86–90 (2013).
55. Mosquera-Lois, I., Klarbring, J. & Walsh, A. Point defect formation at finite temperatures with machine learning force fields. *Chem. Sci.* <https://pubs.rsc.org/en/content/articlelanding/2025/sc/d4sc08582e> (2025).
56. Zhong, X., Höfling, F. & John, T. Hydrogen diffusion in garnet: insights from atomistic simulations. *Geochim. Geophys. Geosyst.* **26**, e2024GC011951 (2025).
57. Mosquera-Lois, I. & Walsh, A. Dynamic vacancy levels in CsPbCl_3 obey equilibrium defect thermodynamics. *PRX Energy* <https://doi.org/10.1103/dxmb-8s96> (2025).
58. Deák, P., Gali, A., Aradi, B. & Frauenheim, T. Accurate gap levels and their role in the reliability of other calculated defect properties. *Phys. Stat. Sol. B* **248**, 790–798 (2011).
59. Lyons, J. L. & Van de Walle, C. G. Computationally predicted energies and properties of defects in GaN. *npj Comput. Mater.* <https://doi.org/10.1038/s41524-017-0014-2> (2017).
60. Gorai, P. & Stevanović, V. Comment on ‘Understanding the intrinsic p-type behavior and phase stability of thermoelectric $\alpha\text{-Mg}_2\text{Sb}_2$ ’. *ACS Appl. Energy Mater.* **3**, 106–108 (2020).
61. Alkauskas, A., Dreyer, C. E., Lyons, J. L. & Van de Walle, C. G. Role of excited states in Shockley–Read–Hall recombination in wide-band-gap semiconductors. *Phys. Rev. B* **93**, 201304 (2016).
62. Kavanagh, S. R. *Accurately Modelling Point Defects in Semiconductors: The Case of CdTe*. Doctoral thesis, Univ. College London (2024).
63. Yang, J.-H., Yin, W.-J., Park, J.-S., Ma, J. & Wei, S.-H. Review on first-principles study of defect properties of CdTe as a solar cell absorber. *Semicond. Sci. Technol.* **31**, 083002 (2016).
64. Pan, J., Metzger, W. K. & Lany, S. Spin–orbit coupling effects on predicting defect properties with hybrid functionals: a case study in CdTe. *Phys. Rev. B* **98**, 054108 (2018).
65. Chatratin, I., Dou, B., Wei, S.-H. & Janotti, A. Doping limits of phosphorus, arsenic, and antimony in CdTe. *J. Phys. Chem. Lett.* **14**, 273–278 (2023).
66. Dabo, I. et al. Koopmans’ condition for density-functional theory. *Phys. Rev. B* <https://doi.org/10.1103/PhysRevB.82.115121> (2010).
67. Demchenko, D. O. et al. Koopmans-tuned Heyd–Scuseria–Ernzerhof hybrid functional calculations of acceptors in GaN. *Phys. Rev. B* <https://doi.org/10.1103/PhysRevB.110.035203> (2024).
68. Chen, W., Griffin, S. M., Rignanes, G.-M. & Hautier, G. Nonunique fraction of Fock exchange for defects in two-dimensional materials. *Phys. Rev. B* **106**, L161107 (2022).
69. Skone, J. H., Govoni, M. & Galli, G. Self-consistent hybrid functional for condensed systems. *Phys. Rev. B* <https://doi.org/10.1103/PhysRevB.89.195112> (2014).
70. Gerosa, M. et al. Defect calculations in semiconductors through a dielectric-dependent hybrid DFT functional: the case of oxygen vacancies in metal oxides. *J. Chem. Phys.* <https://doi.org/10.1063/1.4931805> (2015).
71. Deák, P., Frauenheim, T. & Gali, A. Limits of the scaled shift correction to levels of interstitial defects in semiconductors. *Phys. Rev. B* <https://doi.org/10.1103/PhysRevB.75.153204> (2007).
72. Lany, S. & Zunger, A. Assessment of correction methods for the band-gap problem and for finite-size effects in supercell defect calculations: case studies for ZnO and GaAs. *Phys. Rev. B* **78**, 235104 (2008).
73. Cococcioni, M. & de Gironcoli, S. Linear response approach to the calculation of the effective interaction parameters in the LDA + U method. *Phys. Rev. B* **71**, 035105 (2005).
74. Wang, Z., Brock, C., Matt, A. & Bevan, K. H. Implications of the DFT+U method on polaron properties in energy materials. *Phys. Rev. B* **96**, 125150 (2017).
75. Uhrin, M., Zadoks, A., Binci, L., Marzari, N. & Timrov, I. Machine learning Hubbard parameters with equivariant neural networks. *npj Comput. Mater.* <https://doi.org/10.1038/s41524-024-01501-5> (2025).
76. Timrov, I., Aquilante, F., Cococcioni, M. & Marzari, N. Accurate electronic properties and intercalation voltages of olivine-type Li-ion cathode materials from extended Hubbard functionals. *PRX Energy* <https://doi.org/10.1103/PRXEnergy.1.033003> (2022).
77. Freysoldt, C., Neugebauer, J. & Van de Walle, C. G. Fully *ab initio* finite-size corrections for charged-defect supercell calculations. *Phys. Rev. Lett.* **102**, 016402 (2009).
78. Kumagai, Y. & Oba, F. Electrostatics-based finite-size corrections for first-principles point defect calculations. *Phys. Rev. B* **89**, 195205 (2014).
79. Gake, T., Kumagai, Y., Freysoldt, C. & Oba, F. Finite-size corrections for defect-involving vertical transitions in supercell calculations. *Phys. Rev. B* **101**, 020102 (2020).
80. Kavanagh, S. R. Vaspup2.0. Zenodo (2023).

81. Buckeridge, J., Scanlon, D. O., Walsh, A. & Catlow, C. R. A. Automated procedure to determine the thermodynamic stability of a material and the range of chemical potentials necessary for its formation relative to competing phases and compounds. *Comput. Phys. Commun.* **185**, 330–338 (2014).
82. Stevanović, V., Lany, S., Zhang, X. & Zunger, A. Correcting density functional theory for accurate predictions of compound enthalpies of formation: fitted elemental-phase reference energies. *Phys. Rev. B* **85**, 115104 (2012).
83. Jain, A. et al. Commentary: the materials project: a materials genome approach to accelerating materials innovation. *APL Materials* <https://doi.org/10.1063/1.4812323> (2013).
84. Saal, J. E., Kirklin, S., Aykol, M., Meredig, B. & Wolverton, C. Materials design and discovery with high-throughput density functional theory: the Open Quantum Materials Database (OQMD). *JOM* **65**, 1501–1509 (2013).
85. Kavanagh, S. R. et al. doped: Python toolkit for robust and repeatable charged defect supercell calculations. *J. Open Source Softw.* **9**, 6433 (2024).
86. Zagorac, D., Müller, H., Ruehl, S., Zagorac, J. & Rehme, S. Recent developments in the inorganic crystal structure database: theoretical crystal structure data and related features. *J. Appl. Crystallogr.* **52**, 918–925 (2019).
87. Cen, J., Zhu, B. & Scanlon, D. O. Exploring battery cathode materials in the Li–Ni–O phase diagrams using structure prediction. *J. Phys. Energy* **5**, 035005 (2023).
88. Toriyama, M. Y., Lee, C.-W., Snyder, G. J. & Gorai, P. Defect chemistry and doping of lead phosphate oxo apatite $\text{Pb}_{10}(\text{PO}_4)_6\text{O}$. *ACS Energy Lett.* **9**, 428–431 (2024).
89. Morgan, B. J. & Watson, G. W. Intrinsic *n*-type defect formation in TiO_2 : a comparison of rutile and anatase from GGA+U calculations. *J. Phys. Chem. C* **114**, 2321–2328 (2010).
90. Swallow, J. E. N. et al. Self-compensation in transparent conducting F-doped SnO_2 . *Adv. Funct. Mater.* <https://doi.org/10.1002/adfm.201701900> (2017).
91. Jackson, A. J., Tiana, D. & Walsh, A. A universal chemical potential for sulfur vapours. *Chem. Sci.* **7**, 1082–1092 (2016).
92. Kayastha, P., Longo, G. & Whalley, L. D. A first-principles thermodynamic model for the Ba–Zr–S system in equilibrium with sulfur vapor. *ACS Appl. Energy Mater.* **7**, 11326–11333 (2024).
93. Comparotto, C. et al. Thermodynamic insights into the Ba–S system for the formation of BaZrS_3 perovskites and other Ba sulfides. *J. Mater. Chem. A* **13**, 9983–9991 (2025).
94. Crawford, C. M., Ortiz, B. R., Gorai, P., Stevanovic, V. & Toberer, E. S. Experimental and computational phase boundary mapping of $\text{Co}_4\text{Sn}_6\text{Te}_6$. *J. Mater. Chem. A* **6**, 24175–24185 (2018).
95. Ohno, S. et al. Phase boundary mapping to obtain *n*-type Mg_3Sb_2 -based thermoelectrics. *Joule* **2**, 141–154 (2018).
96. Borgsmiller, L., Zavanelli, D. & Snyder, G. J. Phase-boundary mapping to engineer defects in thermoelectric materials. *PRX Energy* <https://doi.org/10.1103/PRXEnergy.1.022001> (2022).
97. Shimoda, M. et al. Controlling defects to achieve reproducibly high ionic conductivity in Na_3SbS_3 solid electrolytes. *Chem. Mater.* **34**, 5634–5643 (2022).
98. Choi, Y.-S., Costa, S. I. R., Tapia-Ruiz, N. & Scanlon, D. O. Intrinsic defects and their role in the phase transition of Na-ion anode $\text{Na}_7\text{Ti}_3\text{O}_7$. *ACS Appl. Energy Mater.* **6**, 484–495 (2022).
99. Williamson, B. A. D., Eggestad, K. & Selbach, S. M. Pseudo-lithium vacancies in hydrogen rich Li_3OCl . *J. Mater. Chem. A* <https://doi.org/10.1039/d4ta08352k> (2025).
100. Fioretti, A. N. et al. Understanding and control of bipolar self-doping in copper nitride. *J. Appl. Phys.* <https://doi.org/10.1063/1.4948244> (2016).
101. Goyal, A., Gorai, P., Peng, H., Lany, S. & Stevanović, V. A computational framework for automation of point defect calculations. *Comput. Mater. Sci.* **130**, 1–9 (2017).
102. Broberg, D. et al. PyCDT: a Python toolkit for modeling point defects in semiconductors and insulators. *Comput. Phys. Commun.* **226**, 165–179 (2018).
103. Kononov, A., Lee, C.-W., Shapera, E. P. & Schleife, A. Identifying native point defect configurations in α -alumina. *J. Phys. Condens. Matter* **35**, 334002 (2023).
104. Zimmermann, N. E. R., Horton, M. K., Jain, A. & Haranczyk, M. Assessing local structure motifs using order parameters for motif recognition, interstitial identification, and diffusion path characterization. *Front. Mater.* <https://doi.org/10.3389/fmats.2017.00034> (2017).
105. Shen, J.-X. & Varley, J. pymatgen-analysis-defects: a Python package for analyzing point defects in crystalline materials. *J. Open Source Softw.* **9**, 5941 (2024).
106. Ong, S. P. et al. Python Materials Genomics (pymatgen): a robust, open-source Python library for materials analysis. *Comput. Mater. Sci.* **68**, 314–319 (2013).
107. Gorai, P. Beginner's guide to interpreting defect and defect level diagrams. *PRX Energy* **4**, 032001 (2025).
108. Frodason, Y. K., Johansen, K. M., Bjørheim, T. S., Svensson, B. G. & Alkauskas, A. Zn vacancy as a polaronic hole trap in ZnO . *Phys. Rev. B* **95**, 094105 (2017).
109. Lee, C.-W., Din, N. U., Brennecke, G. L. & Gorai, P. Defects and oxygen impurities in ferroelectric wurtzite $\text{Al}_{1-x}\text{Sc}_x\text{N}$ alloys. *Appl. Phys. Lett.* <https://doi.org/10.1063/5.0211892> (2024).
110. Wang, X., Kavanagh, S. R., Scanlon, D. O. & Walsh, A. Four-electron negative-*U* vacancy defects in antimony selenide. *Phys. Rev. B* <https://doi.org/10.1103/PhysRevB.108.134102> (2023).
111. Morris, A. J., Needs, R. J., Salager, E., Grey, C. P. & Pickard, C. J. Lithiation of silicon via lithium zintl-defect complexes from first principles. *Phys. Rev. B* <https://doi.org/10.1103/physrevb.87.174108> (2013).
112. Huang, M. et al. Dasp: defect and dopant ab-initio simulation package. *J. Semicond.* **43**, 042101 (2022).
113. Arrigoni, M. & Madsen, G. K. H. Spinney: post-processing of first-principles calculations of point defects in semiconductors with Python. *Comput. Phys. Commun.* **264**, 107946 (2021).
114. Yuan, Z. & Hautier, G. First-principles study of defects and doping limits in CaO . *Appl. Phys. Lett.* **124**, 232101 (2024).
115. Sokol, A. A. et al. Point defects in ZnO . *Faraday Discuss.* **134**, 267–282 (2007).
116. Kavanagh, S. R., Walsh, A. & Scanlon, D. O. Rapid recombination by cadmium vacancies in CdTe . *ACS Energy Lett.* **6**, 1392–1398 (2021).
117. Zhang, X. et al. Toward a consistent prediction of defect chemistry in CeO_2 . *Chem. Mater.* **35**, 207–227 (2022).
118. Meggiolaro, D. & De Angelis, F. First-principles modeling of defects in lead halide perovskites: best practices and open issues. *ACS Energy Lett.* **3**, 2206–2222 (2018).
119. Swift, M. W., Peelaers, H., Mu, S., Morton, J. J. L. & Van de Walle, C. G. First-principles calculations of hyperfine interaction, binding energy, and quadrupole coupling for shallow donors in silicon. *npj Comput. Mater.* <https://doi.org/10.1038/s41524-020-00448-7> (2020).
120. Kieczka, D. et al. Defects in WS_2 monolayer calculated with a nonlocal functional: any difference from GGA? *Electron. Struct.* **5**, 024001 (2023).
121. Whalley, L. D. Steric engineering of point defects in lead halide perovskites. *J. Phys. Chem. C* **127**, 15738–15746 (2023).
122. Wang, Z., Wei, S. & Zhang, X. Comment on ‘wide-range-tunable p-type conductivity of transparent $\text{Cu}_{1-x}\text{Br}_x$ alloy’. *Adv. Funct. Mater.* <https://doi.org/10.1002/adfm.202112765> (2022).
123. Xu, J. et al. Defect tolerance of mixed b-site organic–inorganic halide perovskites. *ACS Energy Lett.* **6**, 4220–4227 (2021).
124. Peng, H. et al. Convergence of density and hybrid functional defect calculations for compound semiconductors. *Phys. Rev. B* <https://doi.org/10.1103/PhysRevB.88.115201> (2013).
125. Zhang, X., Turiansky, M. E., Shen, J.-X. & Van de Walle, C. G. Defect tolerance in halide perovskites: a first-principles perspective. *J. Appl. Phys.* <https://doi.org/10.1063/5.0083686> (2022).
126. Shi, B. X. et al. General embedded cluster protocol for accurate modeling of oxygen vacancies in metal-oxides. *J. Chem. Phys.* <https://doi.org/10.1063/5.0087031> (2022).
127. Johannes, M. D., Hoang, K., Allen, J. L. & Gaskell, K. Hole polaron formation and migration in olivine phosphate materials. *Phys. Rev. B* <https://doi.org/10.1103/PhysRevB.85.115106> (2012).
128. Hoang, K. & Johannes, M. D. Defect physics in complex energy materials. *J. Phys. Condens. Matter* **30**, 293001 (2018).
129. Woods-Robinson, R. et al. Designing transparent conductors using forbidden optical transitions. *Matter* **6**, 3021–3039 (2023).
130. Oba, F. & Kumagai, Y. Design and exploration of semiconductors from first principles: a review of recent advances. *Appl. Phys. Express* **11**, 060101 (2018).
131. Broberg, D. et al. High-throughput calculations of charged point defect properties with semi-local density functional theory — performance benchmarks for materials screening applications. *npj Comput. Mater.* <https://doi.org/10.1038/s41524-023-01015-6> (2023).
132. Kumagai, Y. et al. Alkali mono-pnictides: a new class of photovoltaic materials by element mutation. *PRX Energy* **2**, 043002 (2023).
133. Deák, P., Udvardhelyi, P., Thiering, G. & Gali, A. The kinetics of carbon pair formation in silicon prohibits reaching thermal equilibrium. *Nat. Commun.* **14**, 361 (2023).
134. Razinkovas, L., Maciaszek, M., Reinhard, F., Doherty, M. W. & Alkauskas, A. Photoionization of negatively charged NV centers in diamond: theory and ab initio calculations. *Phys. Rev. B* **104**, 235301 (2021).
135. Dhaliya, D., Xiong, Y., Sipahigil, A., Griffin, S. M. & Hautier, G. First-principles study of the t center in silicon. *Phys. Rev. Mater.* <https://doi.org/10.1103/PhysRevMaterials.6.L053201> (2022).
136. Dou, W., Spooner, K. B., Kavanagh, S. R., Zhou, M. & Scanlon, D. O. Band degeneracy and anisotropy enhances thermoelectric performance from $\text{Sb}_2\text{S}_3\text{Te}_6$ to $\text{Sb}_2\text{S}_3\text{Te}_8$. *J. Am. Chem. Soc.* **146**, 17679–17690 (2024).
137. Erhart, P., Sadigh, B., Schleife, A. & Åberg, D. First-principles study of codoping in lanthanum bromide. *Phys. Rev. B* <https://doi.org/10.1103/PhysRevB.91.165206> (2015).
138. Larsen, A. H. et al. The atomic simulation environment — a Python library for working with atoms. *J. Phys. Condens. Matter* **29**, 273002 (2017).
139. Kumagai, Y. Corrections on formation energies and eigenvalues of point defect calculations in two-dimensional materials. *Phys. Rev. B* **109**, 054106 (2024).
140. Makov, G. & Payne, M. C. Periodic boundary conditions in ab initio calculations. *Phys. Rev. B* **51**, 4014–4022 (1995).
141. Leslie, M. & Gillan, N. J. The energy and elastic dipole tensor of defects in ionic crystals calculated by the supercell method. *J. Phys. C Solid State Phys.* **18**, 973–982 (1985).
142. Murphy, S. T. & Hine, N. D. M. Anisotropic charge screening and supercell size convergence of defect formation energies. *Phys. Rev. B* **87**, 094111 (2013).
143. Chagas da Silva, M. et al. Self-consistent potential correction for charged periodic systems. *Phys. Rev. Lett.* **126**, 076401 (2021).
144. Suo, Z.-J., Luo, J.-W., Li, S.-S. & Wang, L.-W. Image charge interaction correction in charged-defect calculations. *Phys. Rev. B* **102**, 174110 (2020).
145. Xiao, J. et al. Realistic dimension-independent approach for charged-defect calculations in semiconductors. *Phys. Rev. B* **101**, 165306 (2020).
146. Freysoldt, C. & Neugebauer, J. First-principles calculations for charged defects at surfaces, interfaces, and two-dimensional materials in the presence of electric fields. *Phys. Rev. B* **97**, 205425 (2018).

147. Falletta, S., Wiktor, J. & Pasquarello, A. Finite-size corrections of defect energy levels involving ionic polarization. *Phys. Rev. B* **102**, 041115 (2020).
148. Kumagai, Y. Finite-size corrections to defect energetics along one-dimensional configuration coordinate. *Phys. Rev. B* **107**, L220101 (2023).
149. Freysoldt, C., Neugebauer, J., Tan, A. M. Z. & Hennig, R. G. Limitations of empirical supercell extrapolation for calculations of point defects in bulk, at surfaces, and in two-dimensional materials. *Phys. Rev. B* <https://doi.org/10.1103/PhysRevB.105.014103> (2022).
150. Kokott, S. et al. Efficient all-electron hybrid density functionals for atomistic simulations beyond 10-000 atoms. *J. Chem. Phys.* <https://doi.org/10.1063/5.0208103> (2024).
151. Hayes, W. & Stoneham, A. M. *Defects and Defect Processes in Nonmetallic Solids* (Wiley, 1985).
152. Kasamatsu, S., Tada, T. & Watanabe, S. Theoretical analysis of space charge layer formation at metal/ionic conductor interfaces. *Solid State Ionics* **183**, 20–25 (2011).
153. Neilson, W. D. & Murphy, S. T. DefAP: a Python code for the analysis of point defects in crystalline solids. *Comput. Mater. Sci.* **210**, 111434 (2022).
154. Hagemark, K. Frozen-in native defects in semiconductor compounds. *J. Phys. Chem. Solids* **37**, 461–464 (1976).
155. Sasaki, K. & Maier, J. Low temperature defect chemistry of oxides. *J. Eur. Ceram. Soc.* **19**, 741–745 (1999).
156. Buckeridge, J. Equilibrium point defect and charge carrier concentrations in a material determined through calculation of the self-consistent Fermi energy. *Comput. Phys. Commun.* **244**, 329–342 (2019).
157. Squires, A. G., Scanlon, D. O. & Morgan, B. J. Native defects and their doping response in the lithium solid electrolyte $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$. *Chem. Mater.* **32**, 1876–1886 (2019).
158. Kubicek, M. et al. Oxygen vacancies in fast lithium-ion conducting garnets. *Chem. Mater.* **29**, 7189–7196 (2017).
159. Fallon, M. J. et al. Exploring the anion site disorder kinetics in lithium argyrodites. *J. Am. Chem. Soc.* **147**, 10151–10159 (2025).
160. Eames, C. et al. Ionic transport in hybrid lead iodide perovskite solar cells. *Nat. Commun.* <https://doi.org/10.1038/ncomms8497> (2015).
161. Karmakar, A. et al. Influence of hidden halogen mobility on local structure of $\text{CsSn}(\text{Cl}_{1-x}\text{Br}_x)_3$ mixed-halide perovskites by solid-state NMR. *Chem. Sci.* **12**, 3253–3263 (2021).
162. Chatratin, I., Evangelista, I., McCandless, B., Shafarman, W. & Janotti, A. Defect stability in CdTe based on formation energies and migration barriers. *J. Phys. Chem. C* **129**, 1398–1407 (2025).
163. Ma, J., Yang, J., Wei, S.-H. & Da Silva, J. L. F. Correlation between the electronic structures and diffusion paths of interstitial defects in semiconductors: the case of CdTe. *Phys. Rev. B* **90**, 155208 (2014).
164. Krasikov, D. & Sankin, I. Defect interactions and the role of complexes in the CdTe solar cell absorber. *J. Mater. Chem. A* **5**, 3503–3513 (2017).
165. Squires, A. G., Scanlon, D. O. & Morgan, B. J. Py-sc-fermi: self-consistent Fermi energies and defect concentrations from electronic structure calculations. *J. Open Source Softw.* **8**, 4962 (2023).
166. Arnab, K. A. et al. Quantitative modeling of point defects in $\beta\text{-Ga}_2\text{O}_3$ combining hybrid functional energetics with semiconductor and processes thermodynamics. *Phys. Chem. Chem. Phys.* **27**, 11129–11143 (2025).
167. Kavanagh, S. R., Scanlon, D. O., Walsh, A. & Freysoldt, C. Impact of metastable defect structures on carrier recombination in solar cells. *Faraday Discuss.* **239**, 339–356 (2022).
168. Toriyama, M. Y. et al. How to analyse a density of states. *Mater. Today Electron.* **1**, 100002 (2022).
169. Oliphant, E., Mantena, V., Brod, M., Jeffrey Snyder, G. & Sun, W. Why does silicon have an indirect band gap? *Mater. Horiz.* **12**, 3073–3083 (2025).
170. Fecher, G. H., Kübler, J. & Felser, C. Chirality in the solid state: chiral crystal structures in chiral and achiral space groups. *Materials* **15**, 5812 (2022).
171. Mao, J. et al. High thermoelectric cooling performance of n-type Mg_2Bi_2 -based materials. *Science* **365**, 495–498 (2019).
172. Wang, X., Kavanagh, S. R. & Walsh, A. Sulfur vacancies limit the open-circuit voltage of Sb_2S_3 solar cells. *ACS Energy Lett.* **10**, 161–167 (2025).
173. Liga, S. M., Kavanagh, S. R., Walsh, A., Scanlon, D. O. & Konstantatos, G. Mixed-cation vacancy-ordered perovskites ($\text{Cs}_2\text{Ti}_{1-x}\text{Sn}_x\text{X}_6$; X = I or Br): low-temperature miscibility, additivity, and tunable stability. *J. Phys. Chem. C* **127**, 21399–21409 (2023).
174. Zhang, X., Kang, J. & Wei, S.-H. Defect modeling and control in structurally and compositionally complex materials. *Nat. Comput. Sci.* **3**, 210–220 (2023).
175. Kaewmeechai, C., Strand, J. & Shluger, A. Electronic structure and properties of trapped holes in crystalline and amorphous Ga_2O_3 . *Phys. Rev. B* <https://doi.org/10.1103/PhysRevB.111.035203> (2025).
176. Chen, W., Dahllah, D., Rignanese, G.-M. & Hautier, G. Origin of the low conversion efficiency in $\text{Cu}_2\text{ZnSnS}_4$ kesterite solar cells: the actual role of cation disorder. *Energy Environ. Sci.* **14**, 3567–3578 (2021).
177. Muy, S., Johnston, C. & Marzari, N. AiiDA-defects: an automated and fully reproducible workflow for the complete characterization of defect chemistry in functional materials. *Electron. Struct.* **5**, 024009 (2023).
178. Ogawa, T., Taguchi, A. & Kuwabara, A. An extended computational approach for point-defect equilibria in semiconductor materials. *npj Comput. Mater.* **8**, 79 (2022).
179. Gorai, P. Beginner's guide to interpreting defect and defect level diagrams. *PRX Energy* <https://doi.org/10.1103/jtyg-xry3> (2025).
180. Stevanović, V., Lany, S., Zhang, X. & Zunger, A. Correcting density functional theory for accurate predictions of compound enthalpies of formation: fitted elemental-phase reference energies. *Phys. Rev. B* <https://doi.org/10.1103/PhysRevB.85.115104> (2012).
181. Slavney, A. H., Hu, T., Lindenberg, A. M. & Karunadasa, H. I. A bismuth-halide double perovskite with long carrier recombination lifetime for photovoltaic applications. *J. Am. Chem. Soc.* **138**, 2138–2141 (2016).
182. Sebastião-Luna, P. et al. Tuning the optical absorption of Sn-, Ge-, and Zn-substituted $\text{Cs}_2\text{AgBiBr}_6$ double perovskites: structural and electronic effects. *Chem. Mater.* **33**, 8028–8035 (2021).
183. Lindquist, K. P. et al. Tuning the bandgap of $\text{Cs}_2\text{AgBiBr}_6$ through dilute tin alloying. *Chem. Sci.* **10**, 10620–10628 (2019).
184. Batnaran, B. *Computational Study of the Key Imperfections in Energy Materials*. Master's thesis, Univ. College London (2022).

Acknowledgements

This work was supported by the PRAETORIAN project, funded by UK Research and Innovation (UKRI) under the UK government's Horizon Europe funding guarantee (EP/Y019504/1). The authors thank B. Batnaran for preliminary calculations on the functional dependence of defect structure searches.

Author contributions

S.R.K. researched data for the article, D.O.S. contributed to the discussion of content, A.G.S. wrote the manuscript and all authors reviewed and edited the manuscript before submission.

Competing interests

The authors declare no competing interests.

Additional information

Peer review information *Nature Reviews Materials* thanks Richard Catlow, Prashun Gorai and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© Springer Nature Limited 2026